

Duality, Wasserstein-1, and Semi-discrete OT

Dual potentials, Lipschitz witnesses, Laguerre cells, and quantization algorithms.

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Optimal Transport for Machine Learners

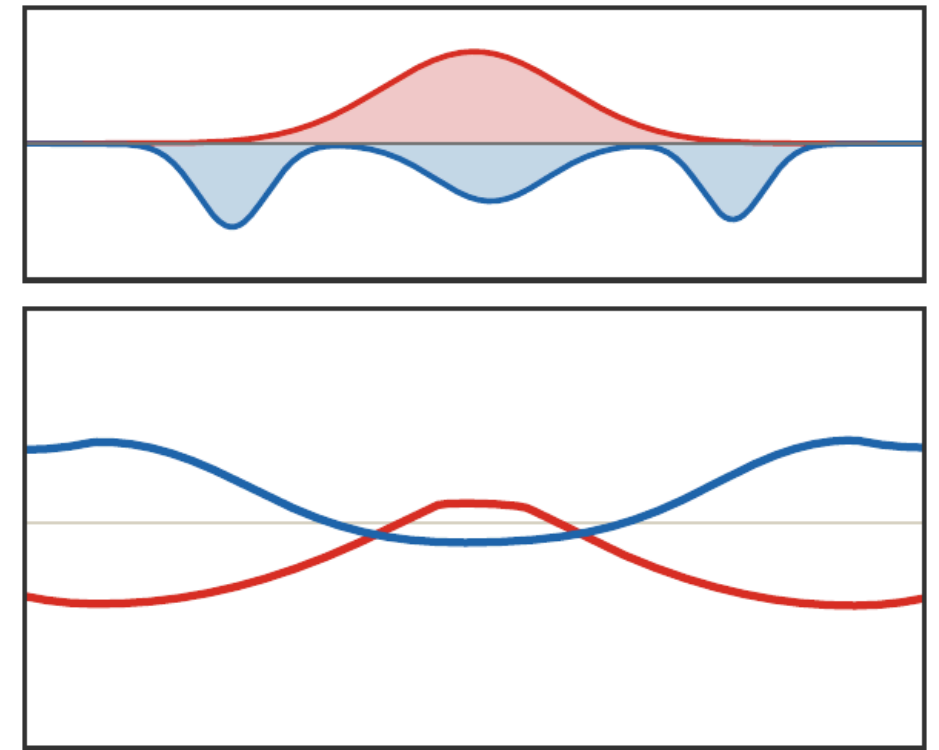
Why duality matters

Kantorovich relaxation turns OT into a linear program. Its dual exposes functions rather than couplings.

This is useful because dual potentials:

- certify optimality,
- define gradients of OT losses,
- produce W_1 witnesses and GAN critics,
- drive semi-discrete algorithms.

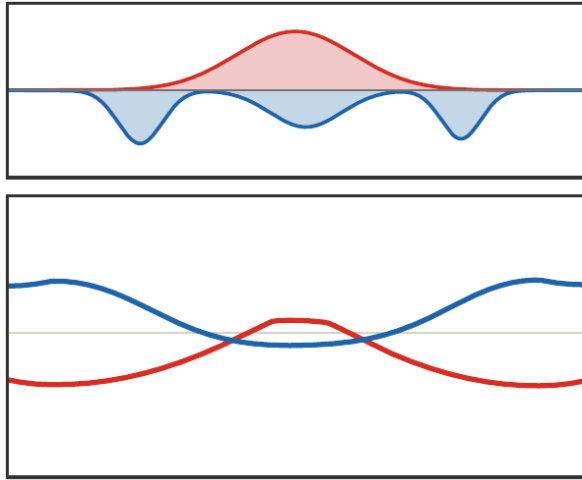
source: α, x, a **target:** β, y, b **plan/map:** π, P, T



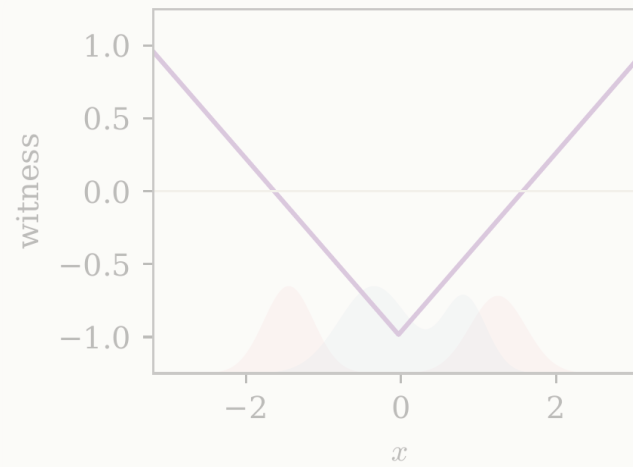
Dual potentials touch the cost on transported pairs.

Roadmap: duality

1. Duality



2. Wasserstein-1 and IPMs



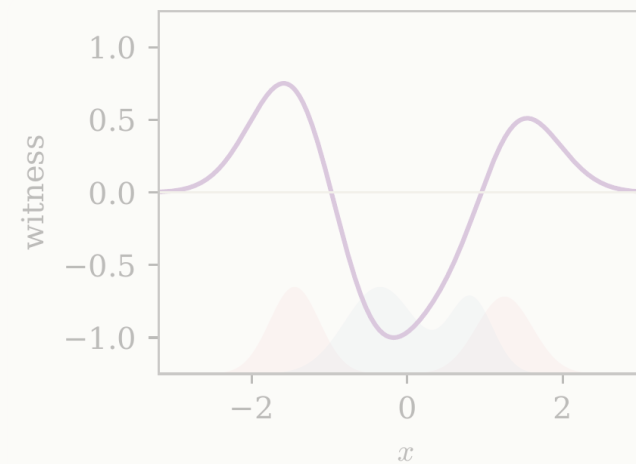
3. Semi-discrete OT



4. Quantization



5. Learning links



Discrete primal and dual

Discrete Kantorovich primal.

For $a \in \Sigma_n$, $b \in \Sigma_m$ and $C \in \mathbb{R}^{n \times m}$,

$$\mathcal{L}_C(a, b) = \min_{P \geq 0} \langle C, P \rangle \quad \text{s.t.} \quad P 1_m = a, \quad P^\top 1_n = b.$$

Discrete dual.

$$\mathcal{L}_C(a, b) = \max_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} \langle f, a \rangle + \langle g, b \rangle \quad \text{s.t.} \quad f_i + g_j \leq C_{i,j}.$$

Lagrangian derivation

The Lagrangian of the primal is

$$\langle C, P \rangle + \langle f, a - P \mathbf{1}_m \rangle + \langle g, b - P^\top \mathbf{1}_n \rangle.$$

Minimizing over $P \geq 0$ gives a finite value if and only if

$$C_{i,j} - f_i - g_j \geq 0 \quad \forall i, j.$$

This yields the dual constraints $f_i + g_j \leq C_{i,j}$.

Complementary slackness

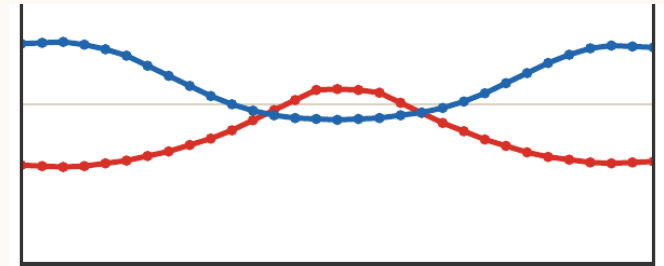
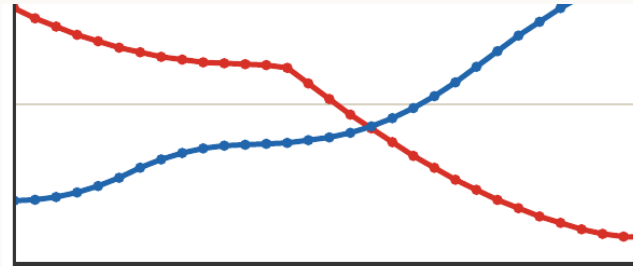
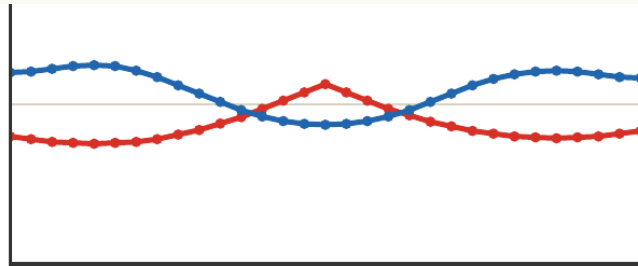
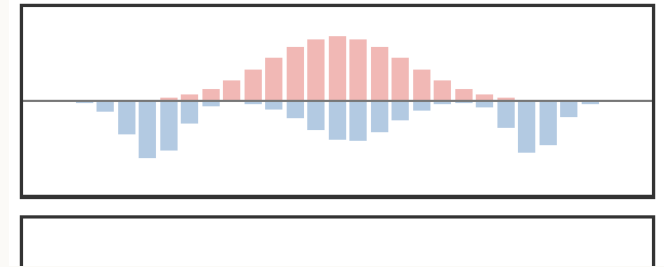
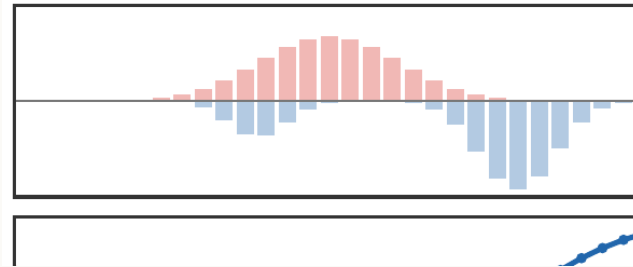
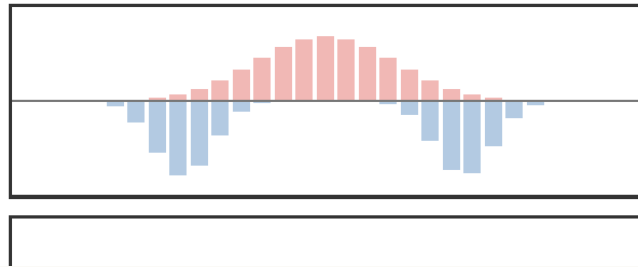
Primal-dual contact.

If P and (f, g) are optimal, then

$$P_{i,j} > 0 \implies f_i + g_j = C_{i,j}.$$

Discrete dual witnesses

Dual variables are not just algebraic multipliers: they are certificates explaining which transport edges can be active.



Live panel: discrete dual potentials



Dual certificates identify the active transport pairs.

Continuous duality

Kantorovich duality.

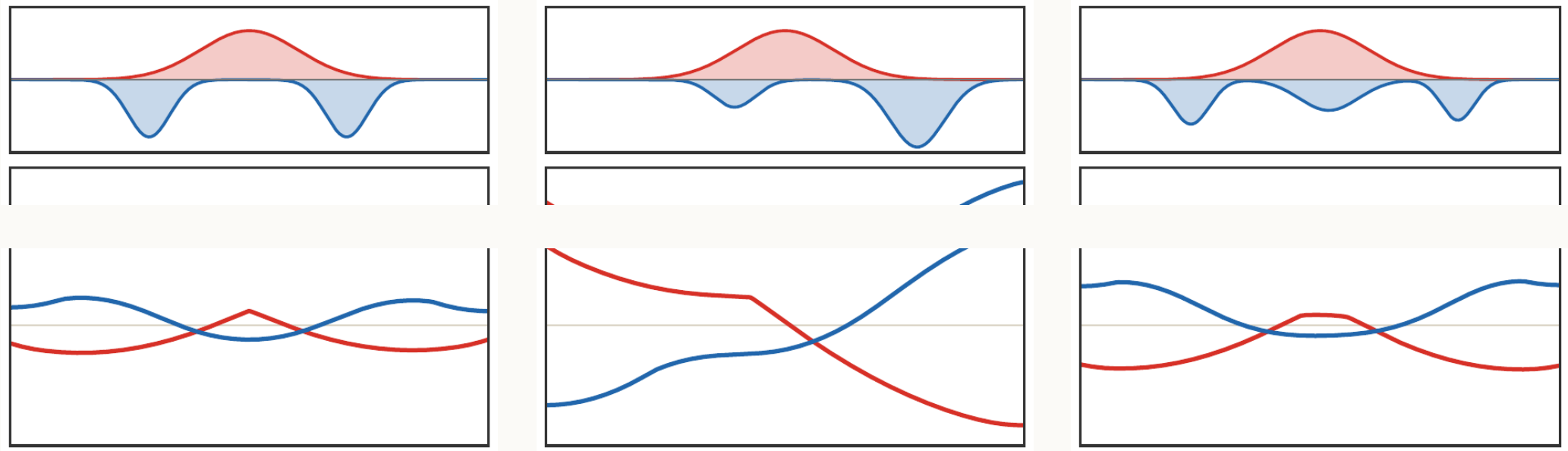
On Polish spaces, for a lower-semicontinuous cost c bounded from below,

$$\mathcal{L}_c(\alpha, \beta) = \sup_{f, g} \int f d\alpha + \int g d\beta \quad \text{s.t.} \quad f(x) + g(y) \leq c(x, y).$$

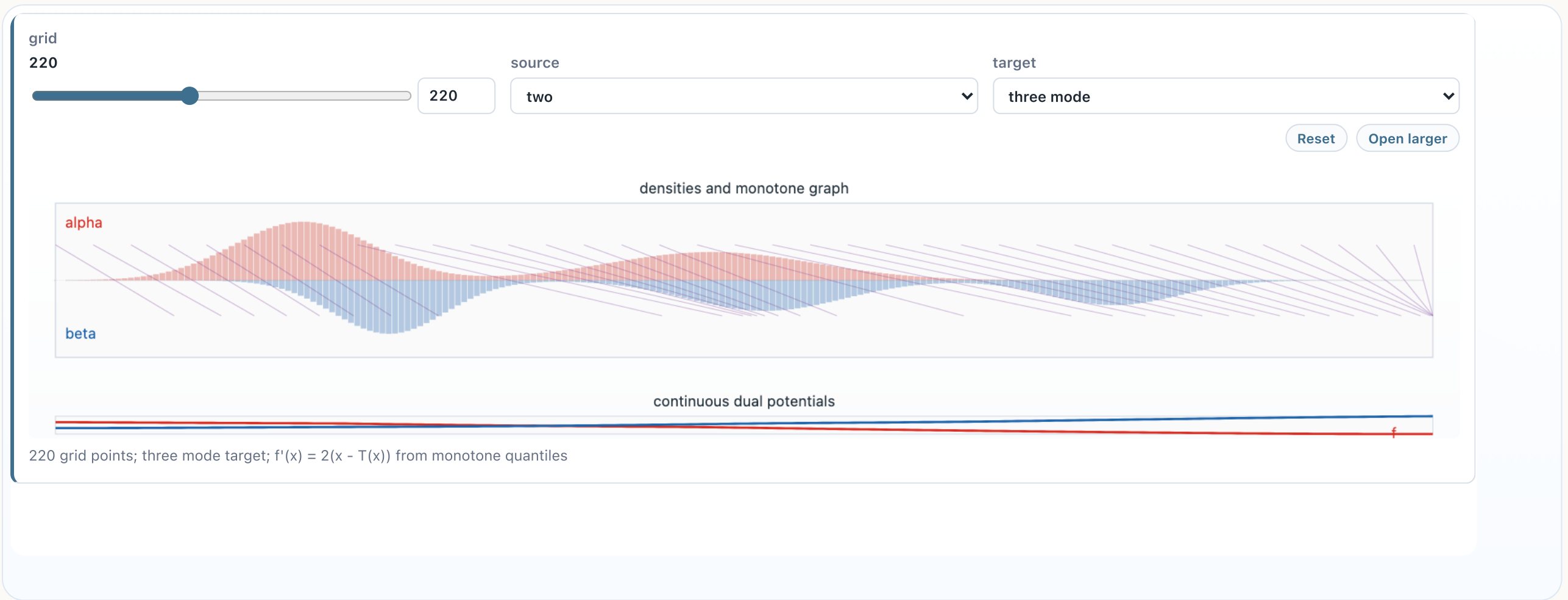
Here “Polish” means complete and separable after choosing a compatible metric. This is the infinite-dimensional analogue of the finite LP dual.

Continuous potentials

For quadratic cost in one dimension, optimal potentials encode the monotone transport map through their derivatives.



Live panel: continuous Kantorovich potentials



In one dimension, the contact condition can be read directly from the potential curves.

c-transform

c-transform.

For a function f on the source space X , define

$$f^c(y) = \inf_{x \in X} (c(x, y) - f(x)).$$

Similarly, for g on the target space Y ,

$$g^{\bar{c}}(x) = \inf_{y \in Y} (c(x, y) - g(y)).$$

The dual can be optimized by replacing one potential by the largest admissible companion.

Reduced dual value

The $\mathcal{C}$ -transform removes one dual variable:

$$\mathcal{L}_c(\alpha, \beta) = \sup_f \left\{ \int f d\alpha + \int f^c d\beta \right\}.$$

In the discrete case this is

$$\mathcal{L}_C(a, b) = \max_{f \in \mathbb{R}^n} \{ \langle f, a \rangle + \langle f^C, b \rangle \}, \quad f_j^C = \min_i (C_{i,j} - f_i).$$

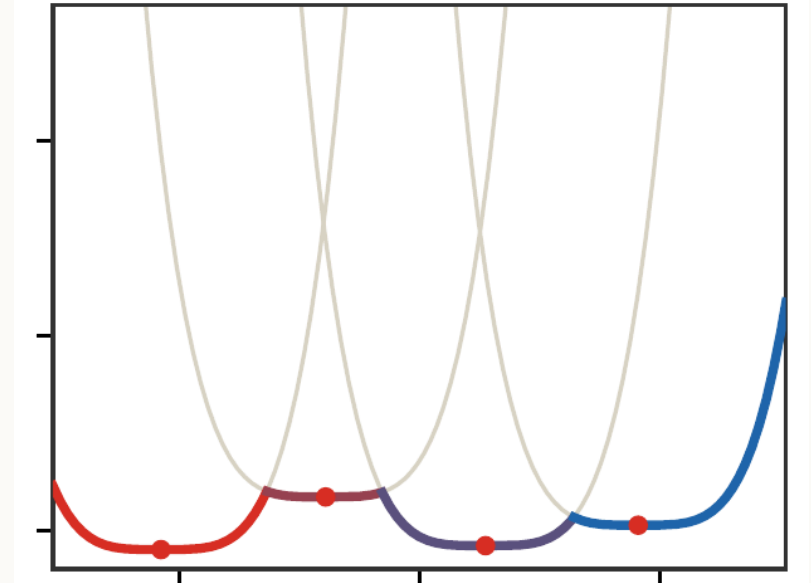
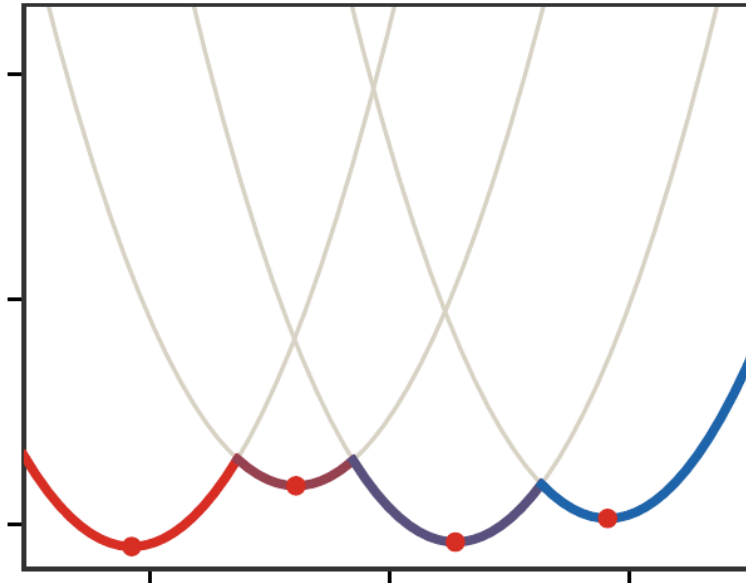
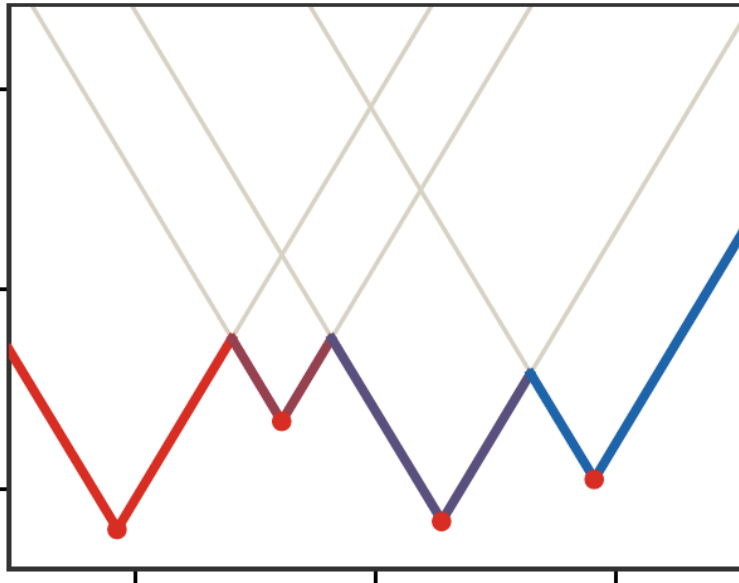
This is the formula behind alternating $\mathcal{C}$ -transform updates.

Reduction to one potential.

At optimum one can impose

$$g = f^c, \quad f = g^{\bar{c}}.$$

Thus dual potentials are fixed points of alternating c-transforms, up to non-uniqueness.



Triple c-transform closure

The identity $f^{ccc} = f^c$ explains why alternating c-transforms stabilize.

Closure identity.

For the c-transform, one always has

$$f^{ccc} = f^c,$$

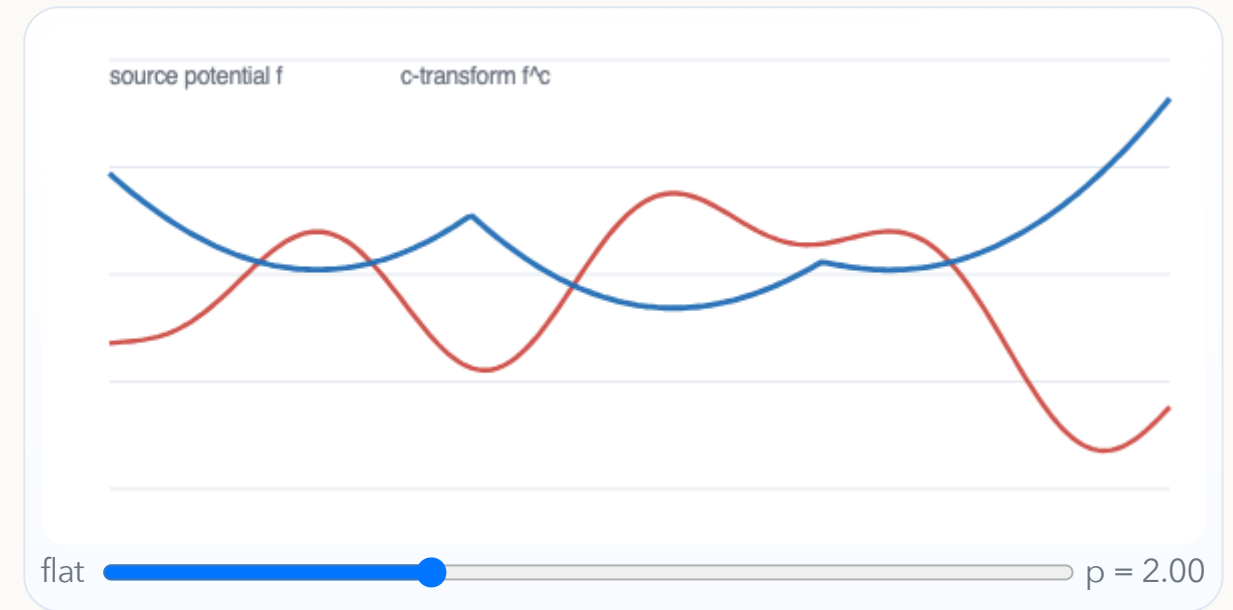
with the appropriate alternation between X and Y .

Proof idea: the inequality $f^{cc} \geq f$ implies $f^{ccc} \leq f^c$, while applying the defining inequality of the c-transform in the reverse direction gives $f^{ccc} \geq f^c$.

Interactive: changing the cost exponent

The c -transform closes a potential under the constraint $f(x) + g(y) \leq c(x, y)$. Changing the cost exponent changes the envelope selected by the dual constraint.

Here $c_p(x, y) = |x - y|^p$ and the slider controls p .



Alternating c-transforms

Alternating minimization in the dual repeatedly closes potentials by c-transforms:

Alternating c-transform closure.

Input: initial source potential f .

Output: dual feasible potentials (f, g) after stabilization.

1. Set an initial source potential f .

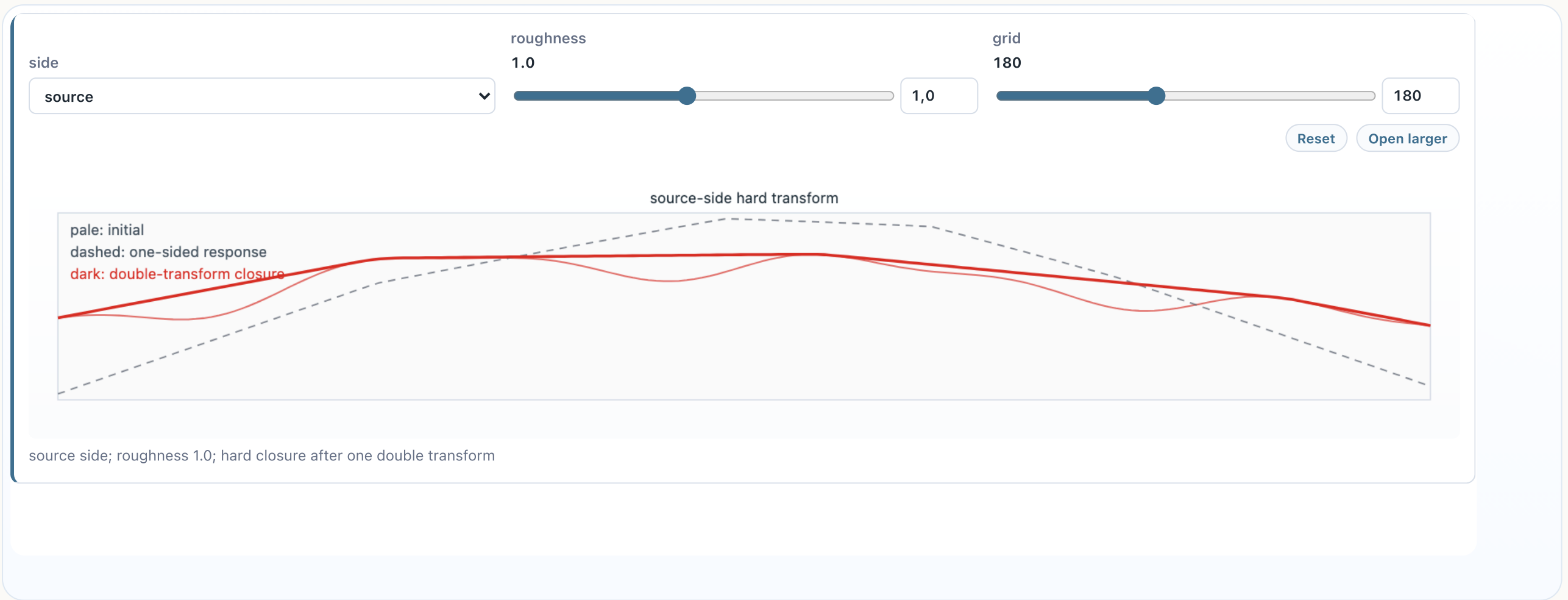
2. Repeat the closure updates

$$g \leftarrow f^c, \quad f \leftarrow g^{\bar{c}}.$$

3. Stop when the potentials are numerically stable.

Useful warning: this reveals the dual structure, but it is not by itself a fast generic OT solver.

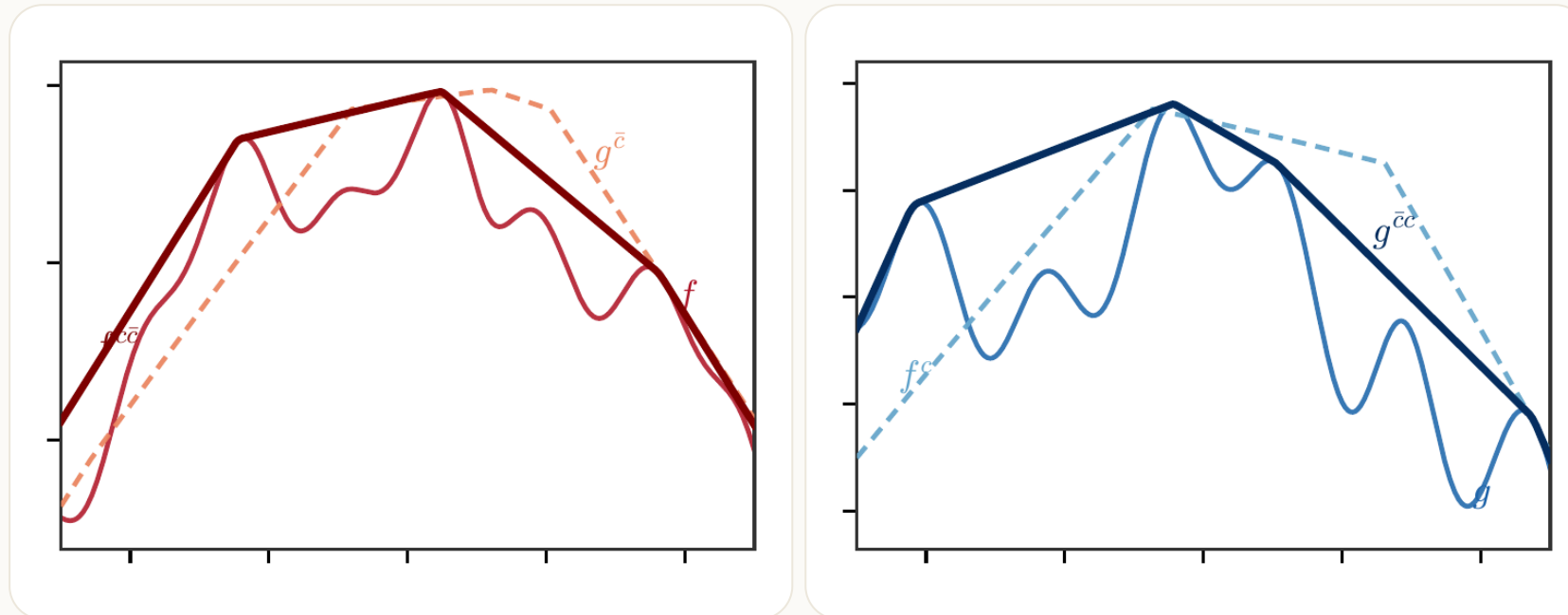
Live panel: alternating c-transforms



Alternating envelopes implement the coordinate ascent geometry of the dual.

c-transform closure geometry

The source and target closures can be viewed as envelopes of admissible potentials.



Live panel: c-transform envelopes



The envelope picture explains why c-concave potentials are the stable objects.

Existence of dual optimizers

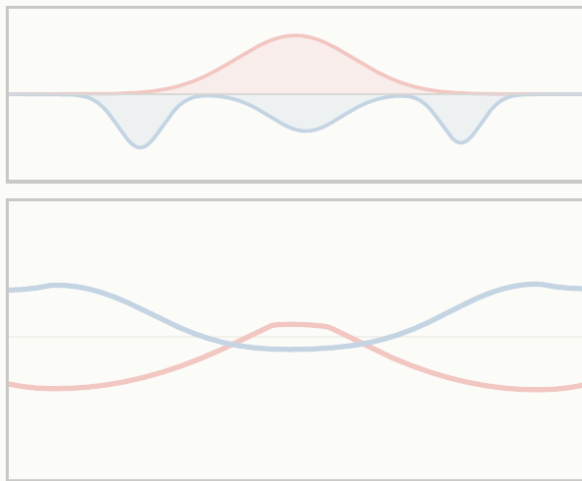
Lipschitz compactness principle.

When the cost is Lipschitz on compact supports, optimal potentials can be chosen uniformly Lipschitz. Arzela-Ascoli compactness then yields an optimizer.

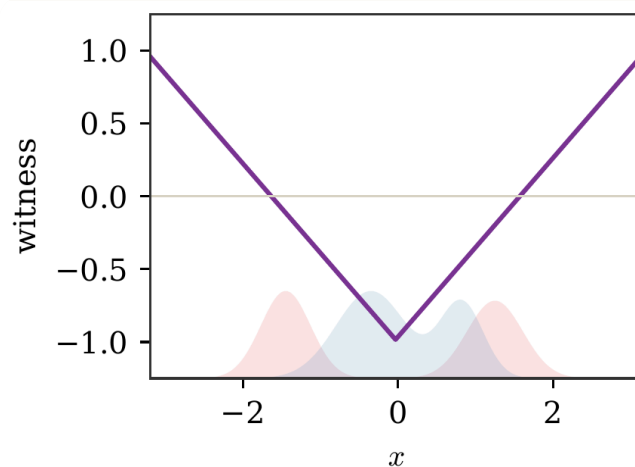
This proof idea is important because the dual problem is over functions, not finite-dimensional vectors.

Roadmap: Wasserstein-1 and integral probability metrics

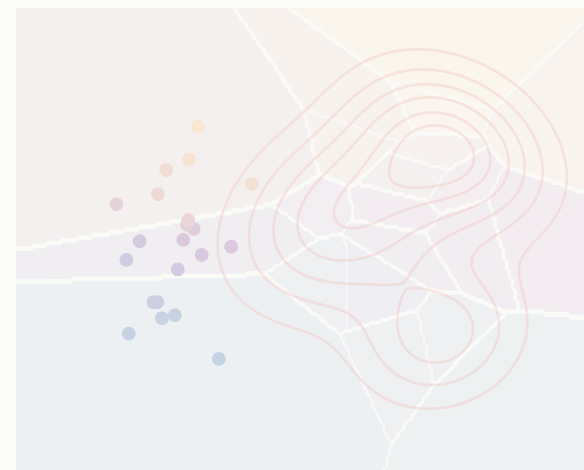
1. Duality



2. Wasserstein-1 and IPMs



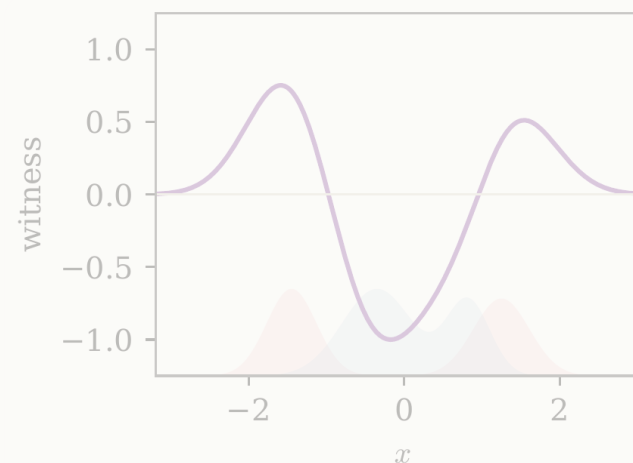
3. Semi-discrete OT



4. Quantization



5. Learning links



OT4ML - Duality and Semi-discrete OT

Kantorovich-Rubinstein formula

For $c(\textcolor{red}{x}, \textcolor{blue}{y}) = d(\textcolor{red}{x}, \textcolor{blue}{y})$, the dual simplifies dramatically.

Wasserstein-1 duality.

$$W_1(\textcolor{red}{\alpha}, \textcolor{blue}{\beta}) = \sup_{\text{Lip}(f) \leq 1} \int f d(\textcolor{red}{\alpha} - \textcolor{blue}{\beta}).$$

Here

$$\text{Lip}(f) = \sup_{\textcolor{red}{x} \neq \textcolor{blue}{y}} \frac{|f(\textcolor{red}{x}) - f(\textcolor{blue}{y})|}{d(\textcolor{red}{x}, \textcolor{blue}{y})}.$$

The witness f is a 1-Lipschitz discriminator between the two measures.

Wasserstein-1 as a norm

Norm viewpoint.

On signed measures with zero total mass,

$$\|\alpha - \beta\|_{W_1} := W_1(\alpha, \beta)$$

is a norm whenever the ground metric is fixed.

This places W_1 among dual norms defined by classes of test functions.

Integral probability metrics

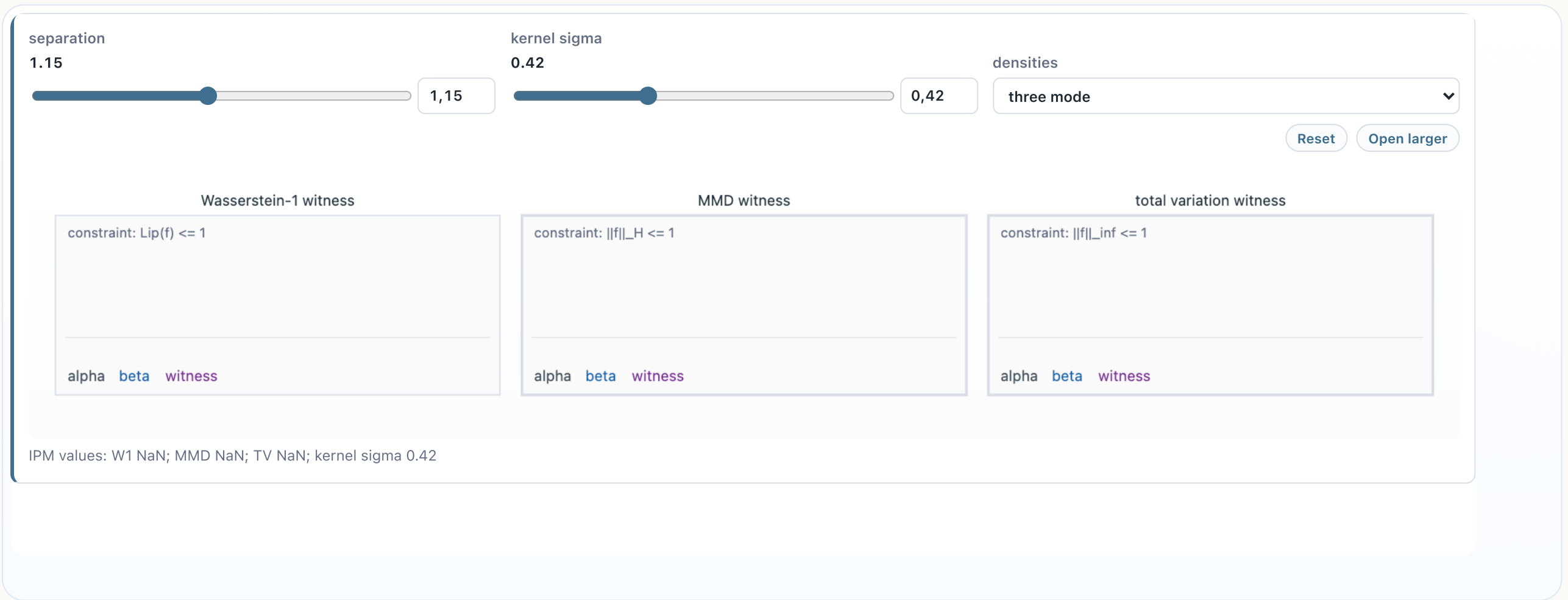
Integral probability metric.

Given a symmetric class of test functions $\mathcal{F}$,

$$D_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \int f d(\alpha - \beta).$$

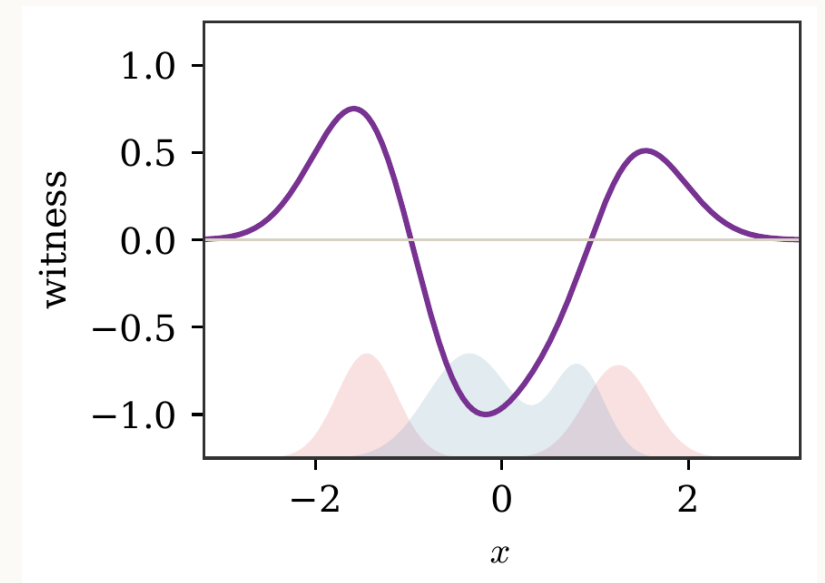
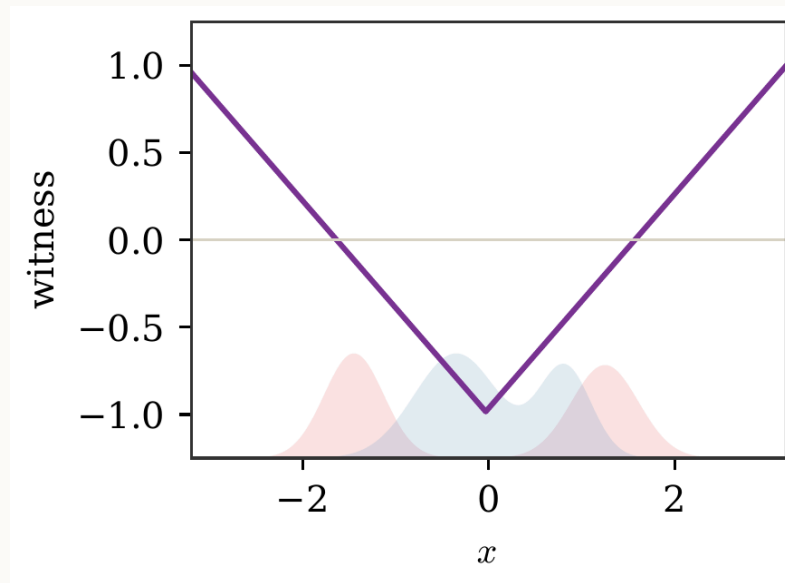
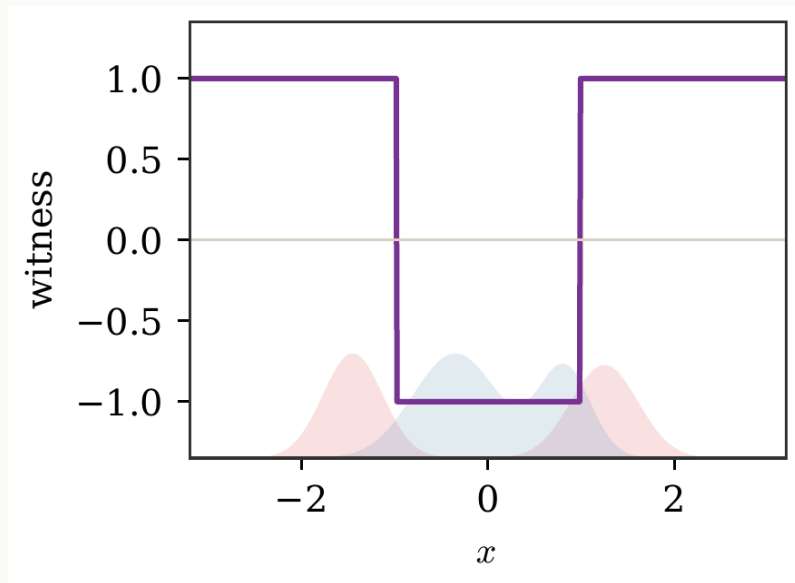
Examples include total variation, Wasserstein-1, bounded-Lipschitz distances, and MMDs.

Live panel: integral probability metrics



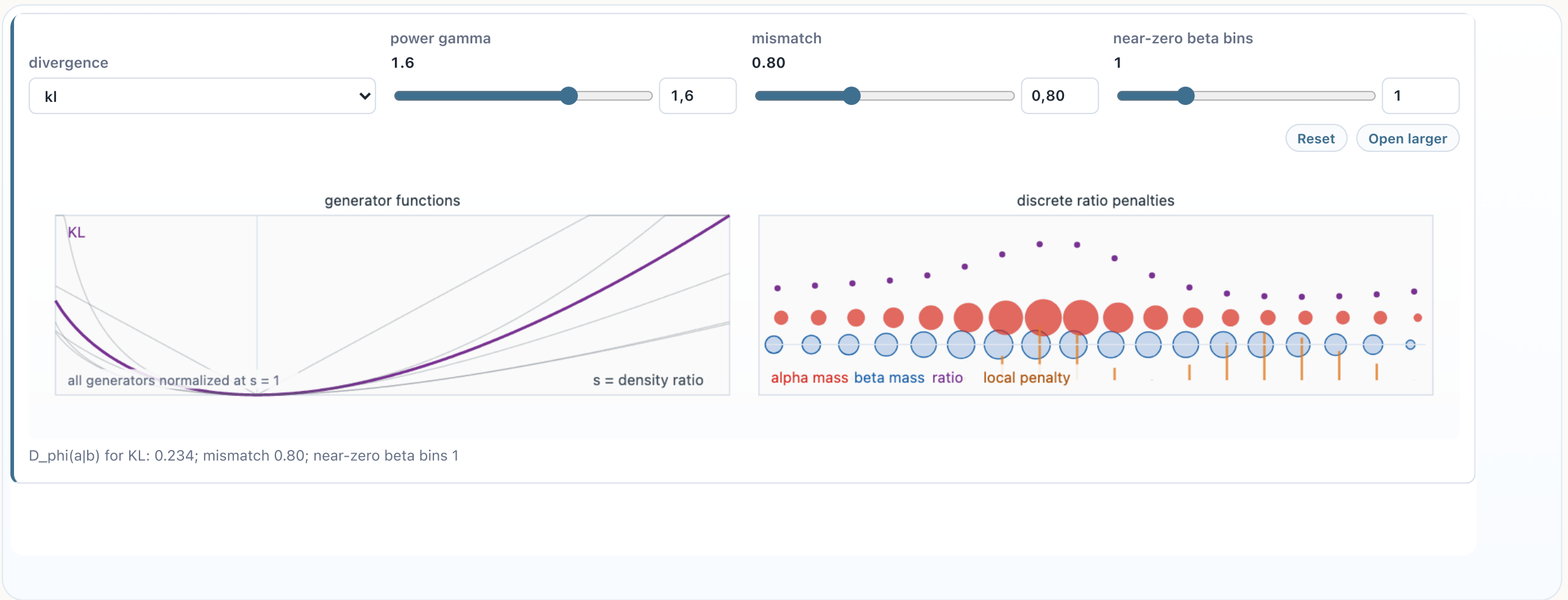
Changing the witness class changes the geometry of the discrepancy.

Dual witnesses compared



Different function classes produce different witnesses and therefore different notions of discrepancy.

Live panel: witness constraints



The same signed mass can be tested by very different admissible witnesses.

Wasserstein-1 on Euclidean domains

For smooth f on Euclidean domains, $\text{Lip}(f) \leq 1$ is equivalent to

$$\|\nabla f(x)\| \leq 1.$$

This connects W_1 to Beckmann's continuous flux formulation:

$$W_1(\alpha, \beta) = \min_m \int \|\mathfrak{m}(x)\| dx \quad \text{s.t.} \quad \text{div } \mathfrak{m} = \alpha - \beta.$$

W1 on surfaces and subdomains

The same Beckmann idea extends beyond flat Euclidean grids by replacing gradients and divergence with their geometric counterparts.

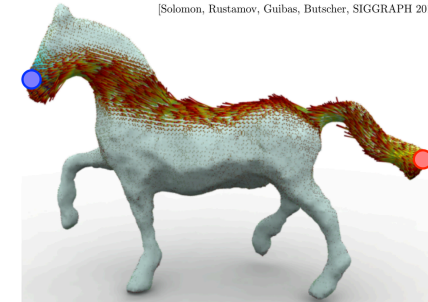
Geometric message.

On meshes, surfaces, or subdomains, W_1 can be computed as a flow constrained by the geometry. Graph discretizations are the simplest finite-dimensional version of this idea.

This is the setting of surface and subdomain W_1 solvers used in geometry processing, including the Solomon-Rustamov-Guibas-Butscher line of work. Graph flows are the simplest discrete proxy for this geometric formulation.

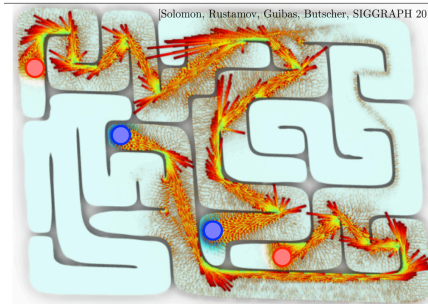
W1 On Surfaces

[Solomon, Rustamov, Guibas, Butscher, SIGGRAPH 2014]



W1 On Sub-domains

[Solomon, Rustamov, Guibas, Butscher, SIGGRAPH 2014]



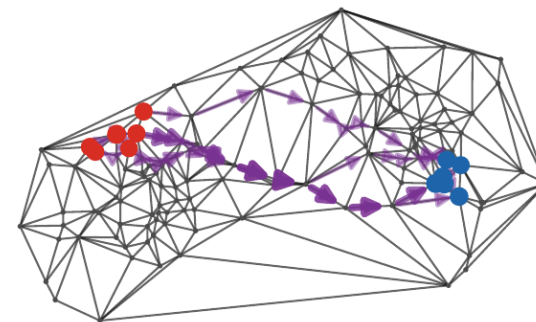
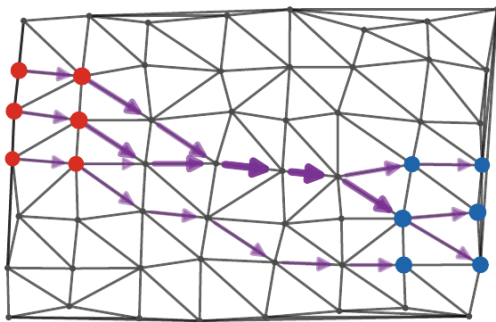
Wasserstein-1 on graphs

On a graph, the Beckmann formulation becomes a sparse network-flow problem over edges:

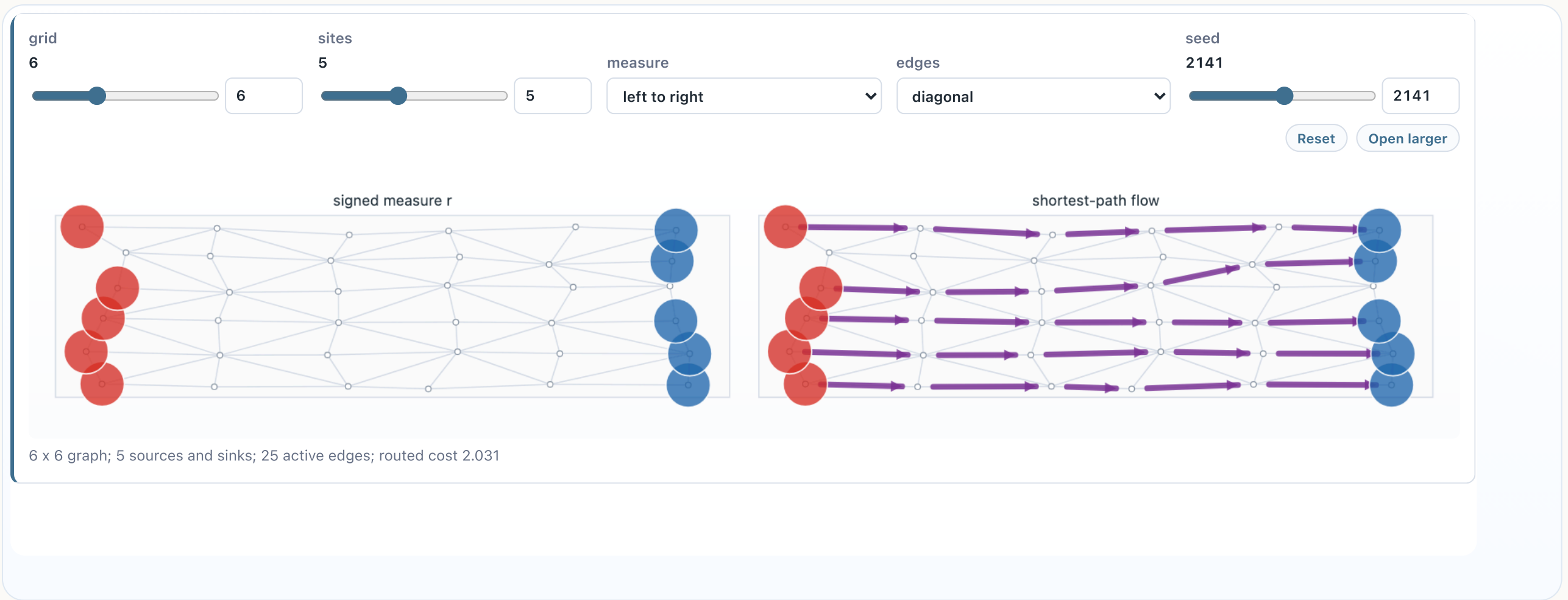
$$\min_q \sum_e w_e |q_e| \quad \text{s.t.} \quad Dq = a - b.$$

Here D is the signed vertex-edge incidence matrix and q_e is the oriented mass flux on edge e .

This can be much smaller than a full pairwise Kantorovich LP.

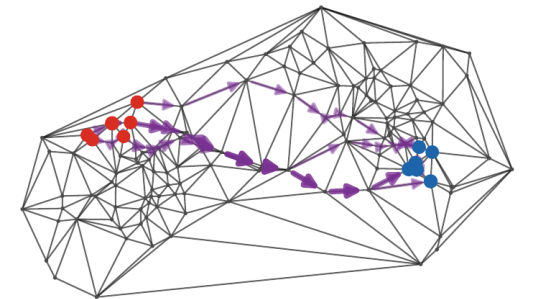
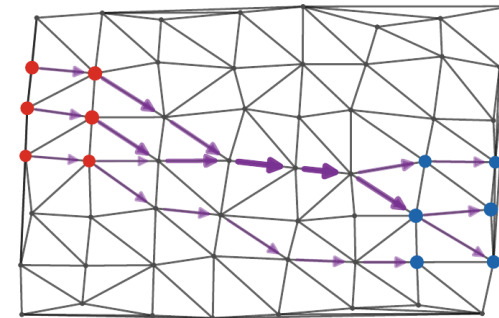
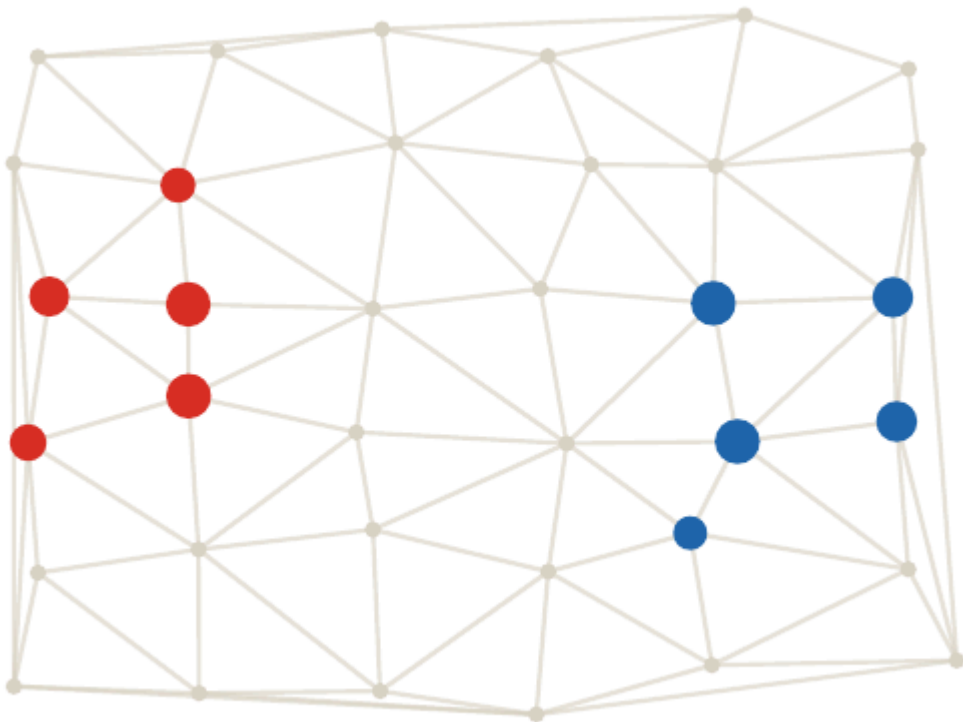


Live panel: graph Beckmann transport



The graph formulation turns W_1 into a network flow problem.

Graph flow visualization



The edge flow is the sparse object that transports positive mass to negative mass through the graph geometry.

WGAN viewpoint

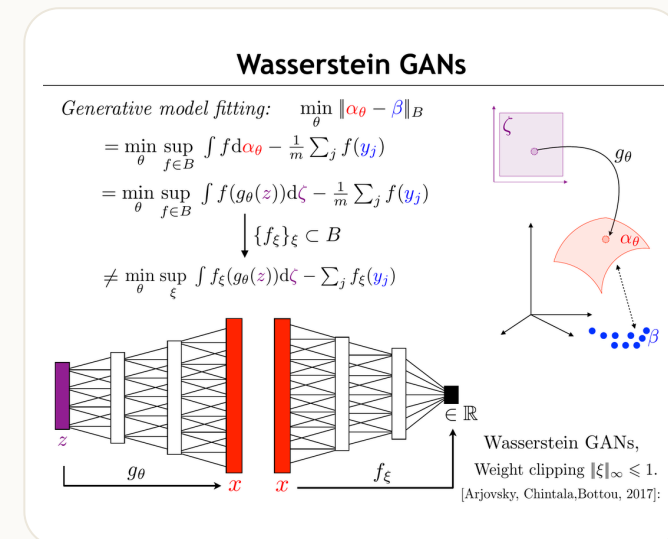
Example: Wasserstein GAN.

A generator g_θ can be trained by minimizing

$$\min_{\theta} W_1((g_\theta)_\# \zeta, \beta) = \min_{\theta} \sup_{\text{Lip}(f) \leq 1} \mathbb{E} f(g_\theta(Z)) - \mathbb{E} f(Y).$$

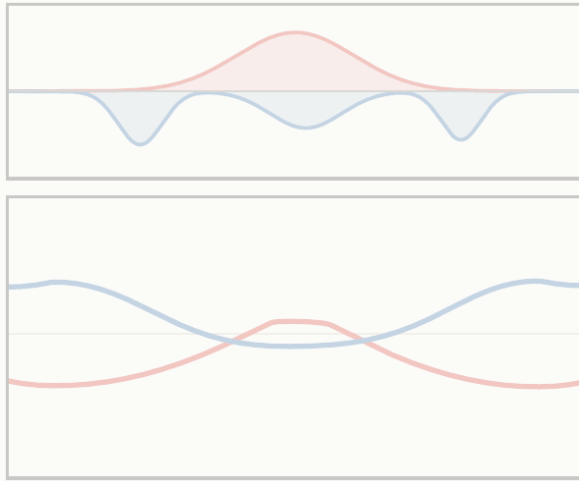
The discriminator is a parametrized approximation of a 1-Lipschitz Kantorovich potential.

This is the machine-learning face of Kantorovich-Rubinstein duality.

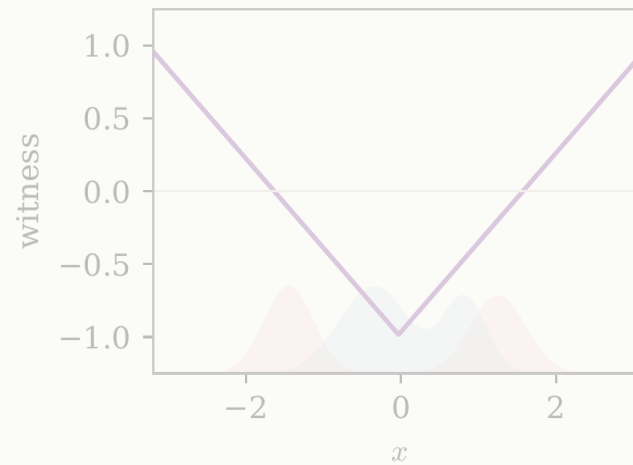


Roadmap: semi-discrete OT

1. Duality



2. Wasserstein-1 and IPMs



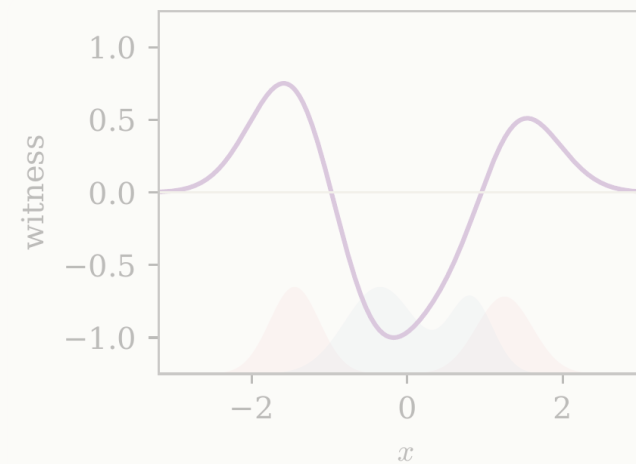
3. Semi-discrete OT



4. Quantization



5. Learning links



Semi-discrete setting

Semi-discrete OT.

Let α be a continuous probability measure on $\mathbb{R}^d$ and

$$\beta = \sum_{j=1}^m b_j \delta_{y_j}$$

be discrete. For quadratic cost, one seeks a partition of the **source** domain whose cells have masses b_j .

The unknown is not a full coupling matrix but a weighted spatial partition.

Laguerre cells

Laguerre cell.

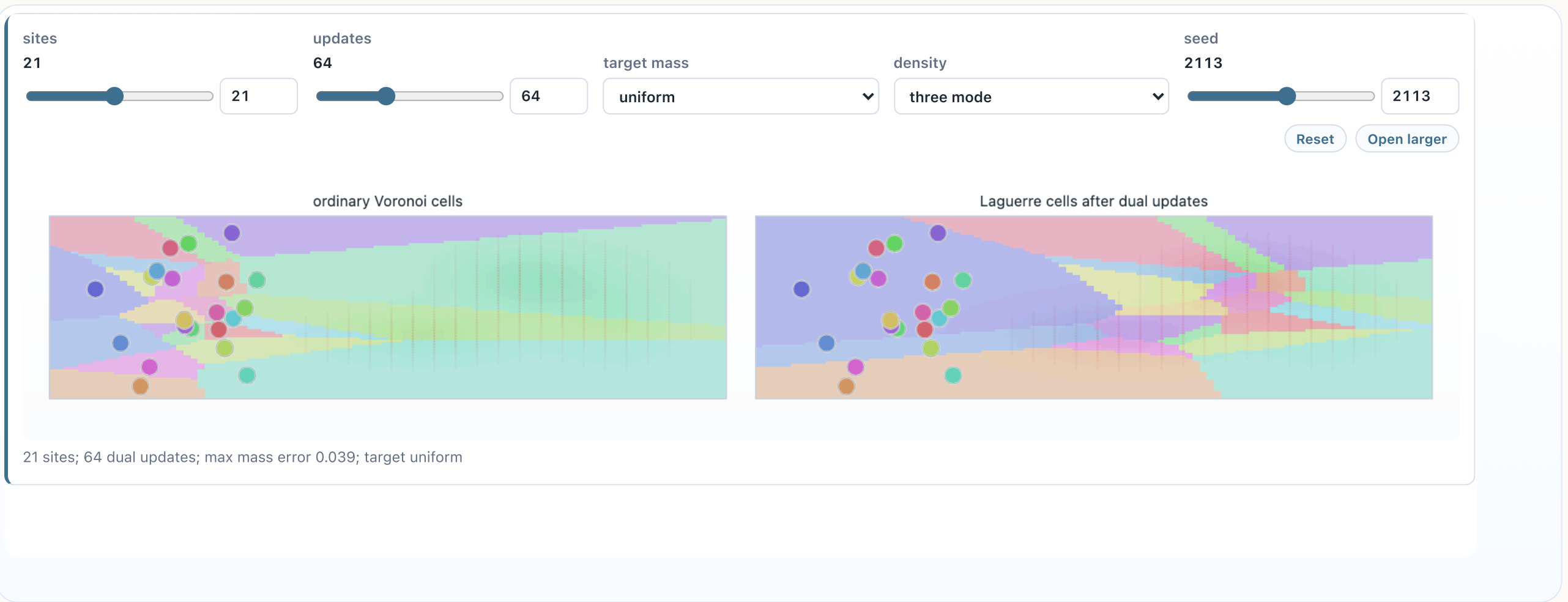
For target sites y_j and weights g_j , the cell associated with j is

$$L_j(g) = \{x : \|x - y_j\|^2 - g_j \leq \|x - y_k\|^2 - g_k \quad \forall k\}.$$

The semi-discrete **map** sends every $x \in L_j(g)$ to y_j . Optimal weights are characterized by the mass constraints

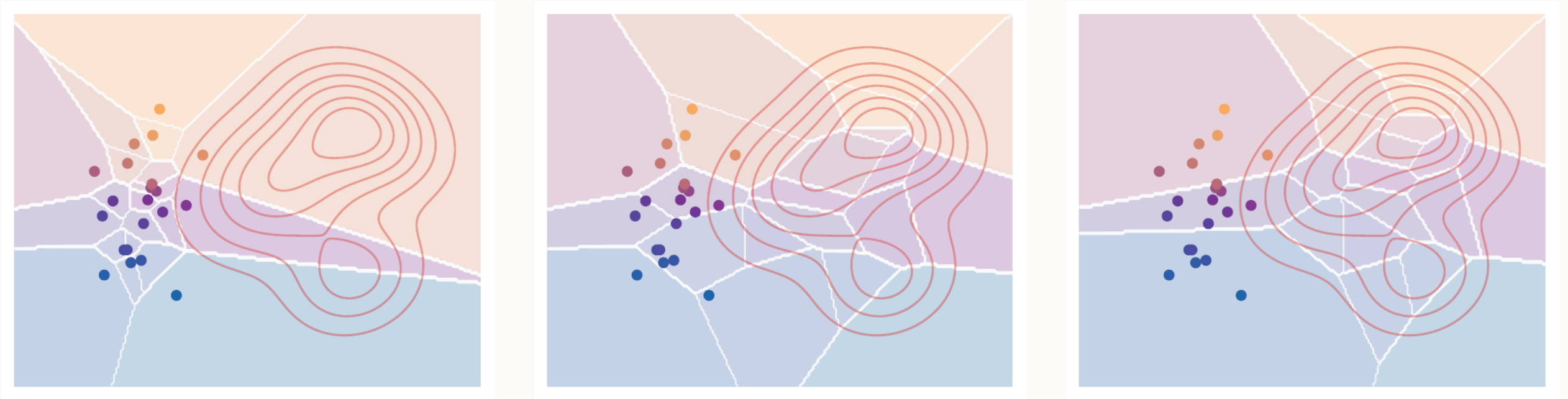
$$\alpha(L_j(g)) = b_j, \quad j = 1, \dots, m.$$

Live panel: Laguerre cells and dual weights



Changing a target weight geometrically expands or shrinks its cell.

Power diagrams



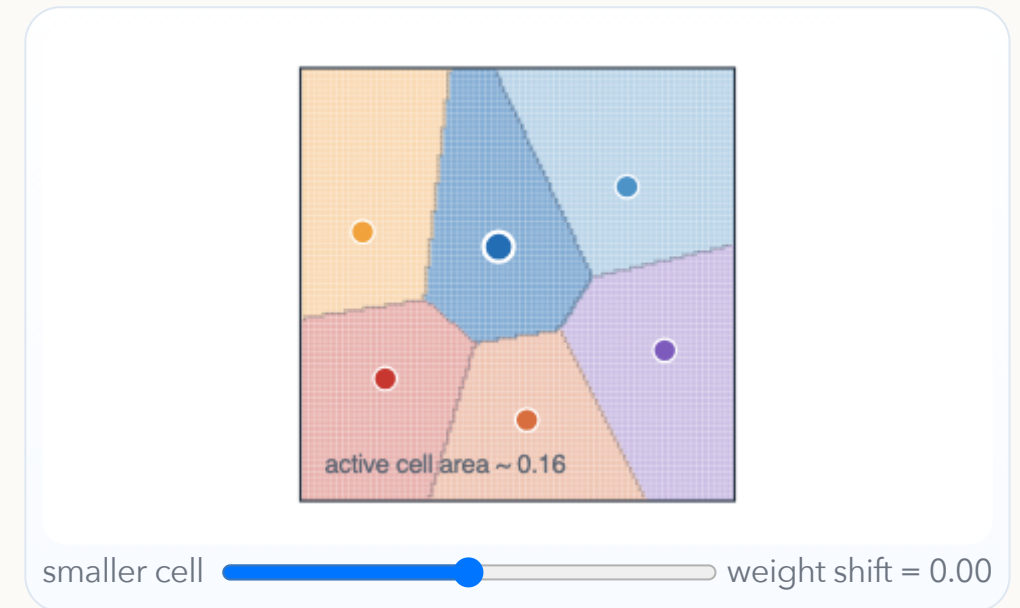
Voronoi cells become Laguerre cells when weights are introduced. The weights tune cell masses.

Interactive: one weight changes one cell

Changing g_j shifts the additive power distance

$$\|\mathbf{x} - \mathbf{y}_j\|^2 - g_j.$$

Increasing g_j makes site j more attractive and expands its Laguerre cell.



Semi-discrete dual

For quadratic cost, the dual can be written as a concave maximization over weights:

$$\max_{g \in \mathbb{R}^m} \int \min_j (\|x - y_j\|^2 - g_j) d\alpha(x) + \sum_{j=1}^m b_j g_j.$$

Gradient.

The gradient component is

$$\frac{\partial}{\partial g_j} = b_j - \alpha(L_j(g)).$$

Thus dual ascent balances Laguerre cell masses.

Hessian and Newton structure

The reference semi-discrete solver uses the fact that the dual has an explicit sparse Hessian. If $\alpha = \rho(x) dx$ and cells $L_i(g)$ and $L_j(g)$ share a facet

$$F_{i,j} = L_i(g) \cap L_j(g),$$

then for $i \neq j$,

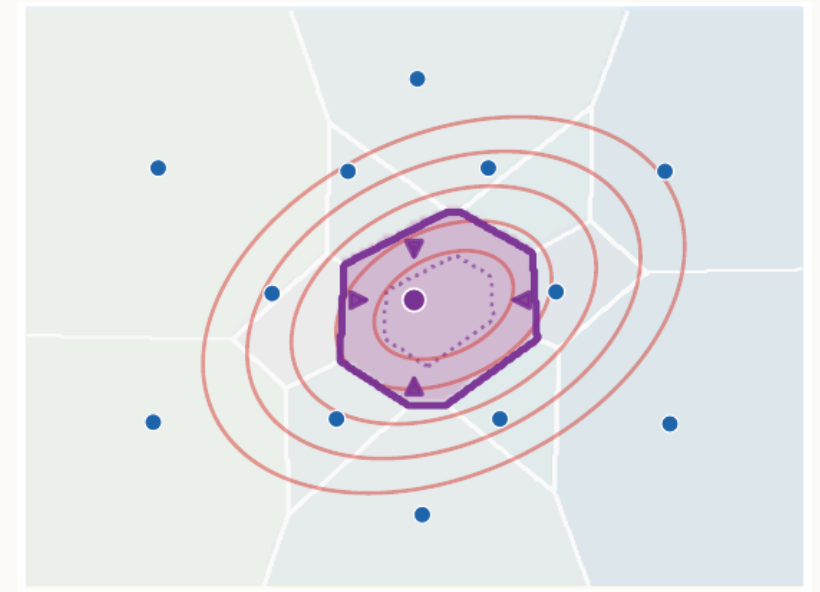
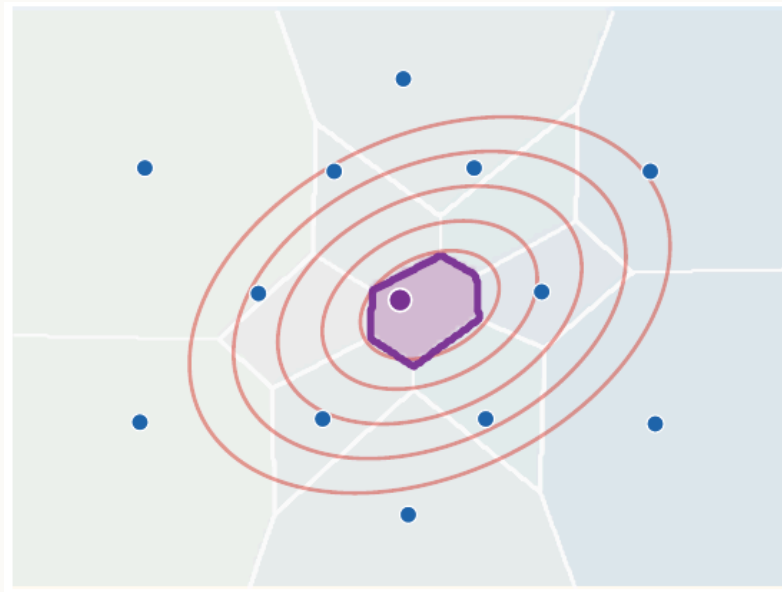
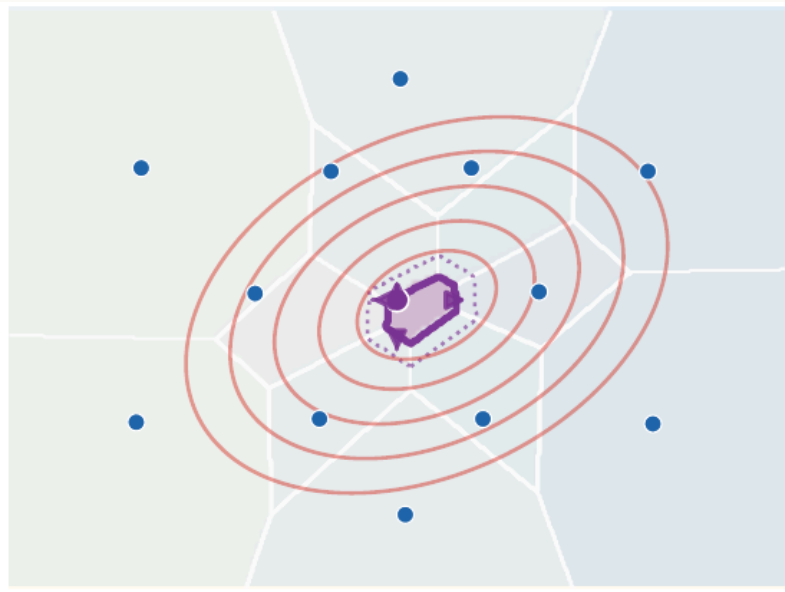
$$\frac{\partial \alpha(L_i(g))}{\partial g_j} = -\frac{1}{2\|y_i - y_j\|} \int_{F_{i,j}} \rho(x) d\mathcal{H}^{d-1}(x),$$

and the diagonal is chosen so rows sum to zero:

$$\frac{\partial \alpha(L_i(g))}{\partial g_i} = -\sum_{j \neq i} \frac{\partial \alpha(L_i(g))}{\partial g_j}.$$

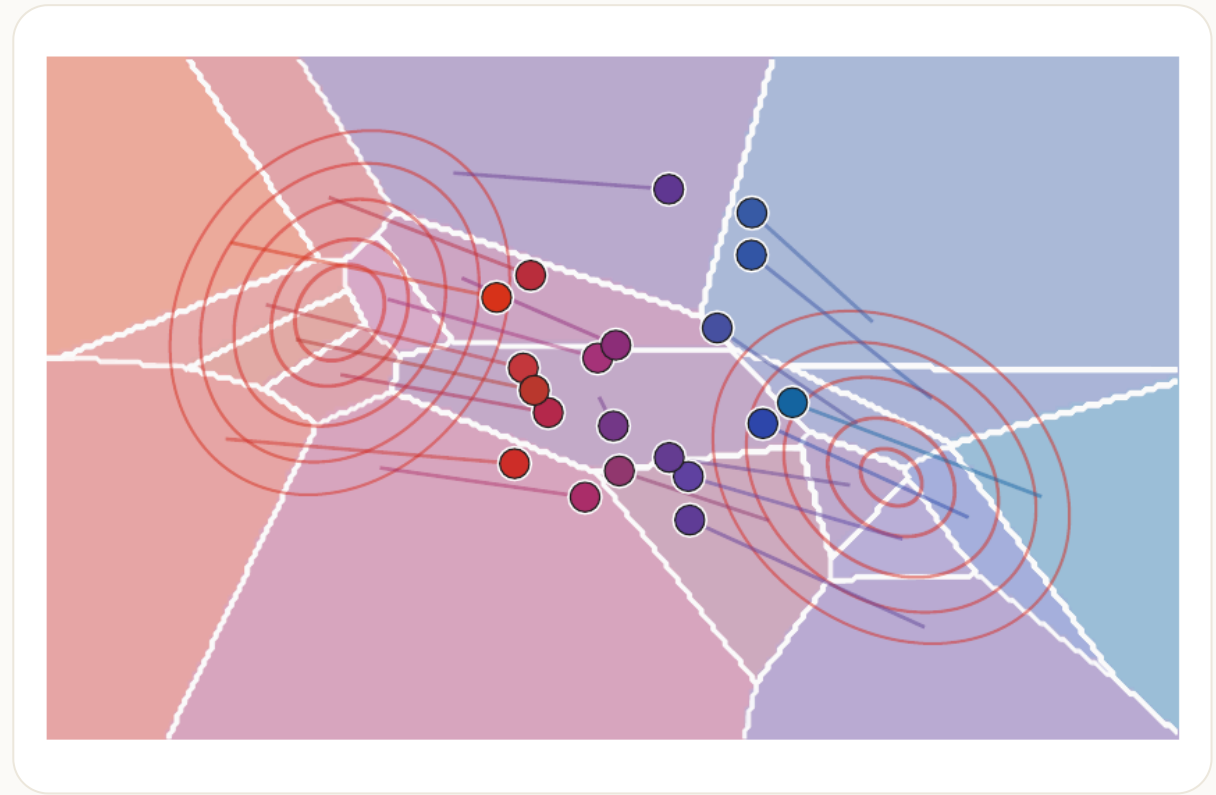
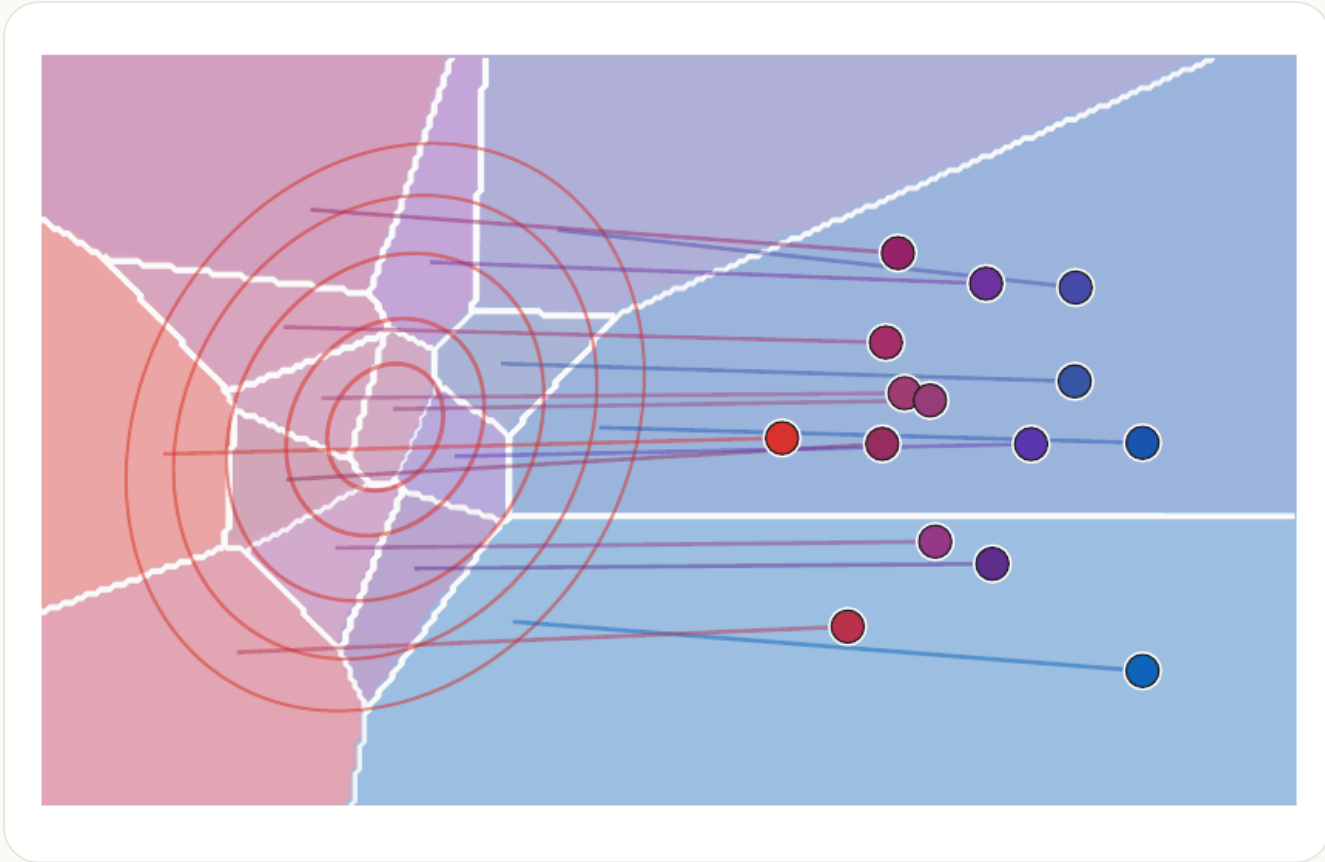
Thus the Hessian of the concave dual has positive off-diagonal entries and negative diagonal row sums. Newton or quasi-Newton methods exploit the adjacency graph of the Laguerre diagram rather than a dense coupling matrix.

Weight descent as mass balancing



The algorithm updates the weights until every cell has the prescribed mass.

Semi-discrete examples



Coloring makes the spatial map visible: each source cell is sent to one target atom.

Practical optimization

Semi-discrete dual ascent.

Input: source density α , target sites y_j , target masses b_j .

Initialize weights g .

Repeat:

$$L_j(g) = \{x : \|x - y_j\|^2 - g_j \leq \|x - y_k\|^2 - g_k\},$$

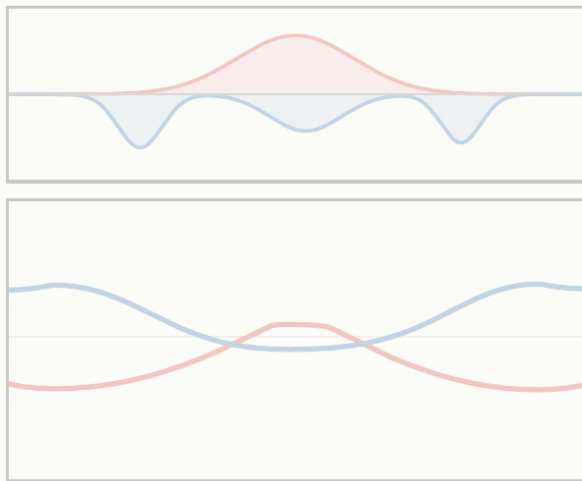
$$g_j \leftarrow g_j + \tau(b_j - \alpha(L_j(g))).$$

Output: the Laguerre partition and map $T(x) = y_j$ for $x \in L_j(g)$.

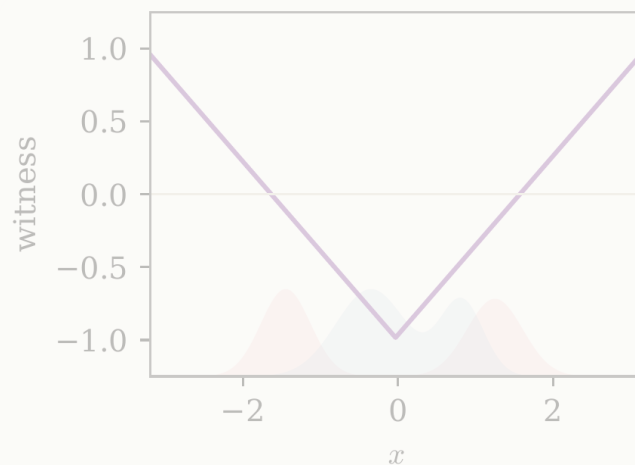
Newton and quasi-Newton methods, especially L-BFGS variants, are typically used in high-quality implementations because the semi-discrete dual is a smooth concave objective away from cell degeneracies.

Roadmap: optimal quantization

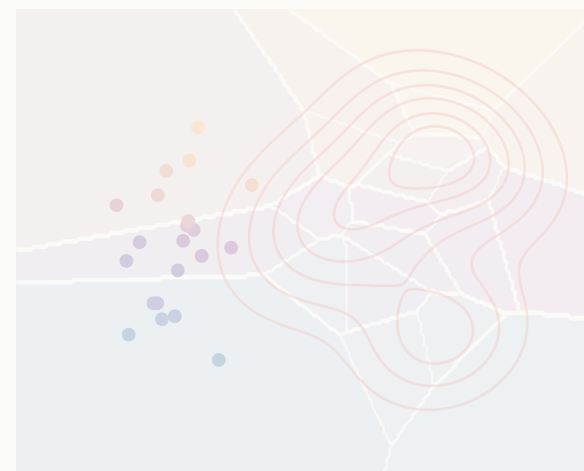
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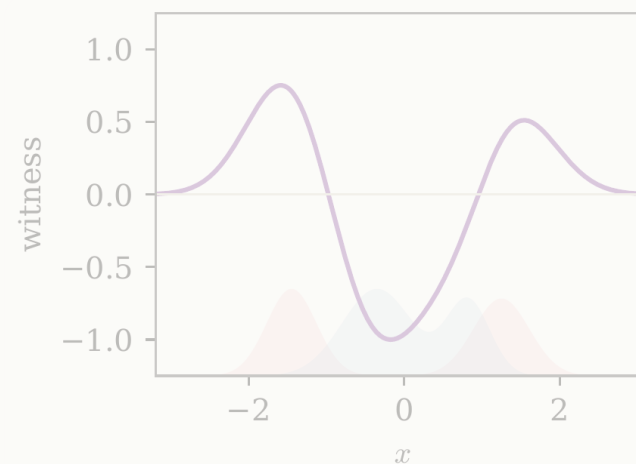
3. Semi-discrete OT



4. Quantization



5. Learning links



OT4ML - Duality and Semi-discrete OT

Quantization problem

Optimal quantization.

For a source measure α and m target sites $Y = (y_1, \dots, y_m)$,

$$Q_m(\alpha) = \min_{Y, b \in \Sigma_m} W_2^2 \left(\alpha, \sum_{j=1}^m b_j \delta_{y_j} \right).$$

For fixed sites Y , this equals

$$\int \min_j \|x - y_j\|^2 d\alpha(x)$$

when the masses are free.

The goal is to approximate a continuous measure by a small discrete measure.

Fixed versus free masses

If the masses b_j are fixed, semi-discrete OT gives the optimal cells for fixed sites. If the masses are free, the optimal cells are Voronoi cells.

Centroid condition.

At an unconstrained local optimum for quadratic cost,

$$y_j = \frac{\int_{V_j(Y)} x d\alpha(x)}{\alpha(V_j(Y))},$$

where $V_j(Y)$ is the Voronoi cell of y_j .

Quantization stationarity

For fixed sites Y and free masses, the semi-discrete dual collapses to the closest-site formula

$$\min_{b \in \Sigma_m} \mathcal{L}_c \left(\alpha, \sum_j b_j \delta_{y_j} \right) = \int \min_j c(x, y_j) d\alpha(x).$$

If the cells $V_j(Y) = \{x : c(x, y_j) \leq c(x, y_k) \ \forall k\}$ are stable and c is differentiable in y , a local minimizer satisfies

$$\int_{V_j(Y)} \nabla_y c(x, y_j) d\alpha(x) = 0.$$

For $c(x, y) = \|x - y\|^2$, this is exactly the centroid condition.

Lloyd algorithm

Lloyd iteration.

Input: source measure α , initial sites Y .

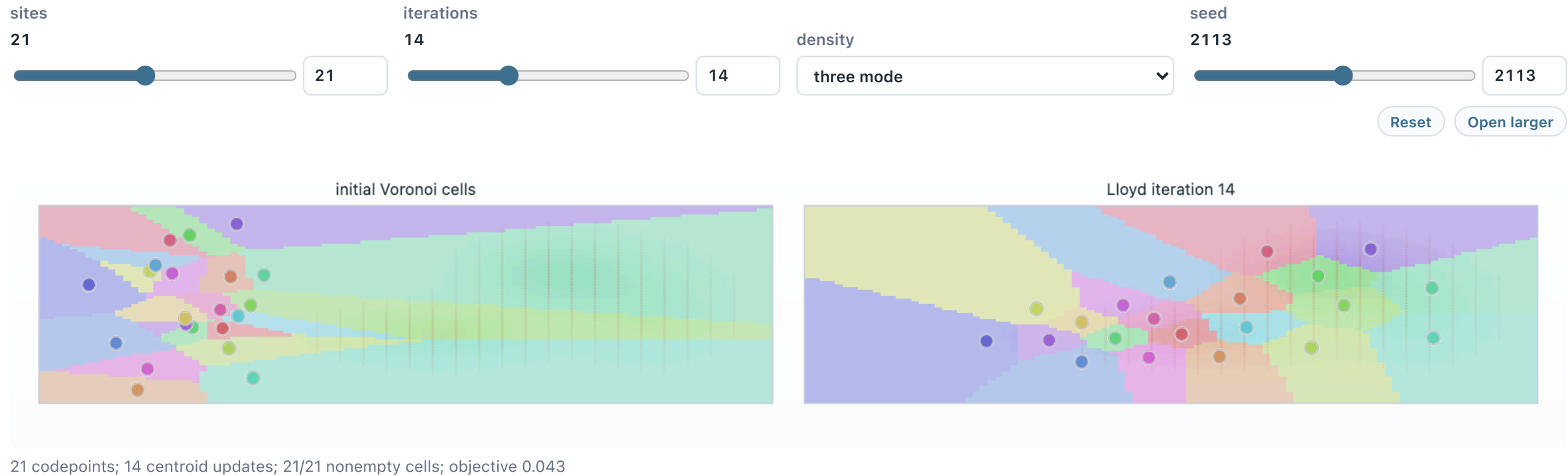
Repeat:

1. Build Voronoi cells $V_j(Y)$.
2. Replace each site by its cell centroid,

$$y_j \leftarrow \frac{\int_{V_j(Y)} x d\alpha(x)}{\alpha(V_j(Y))}.$$

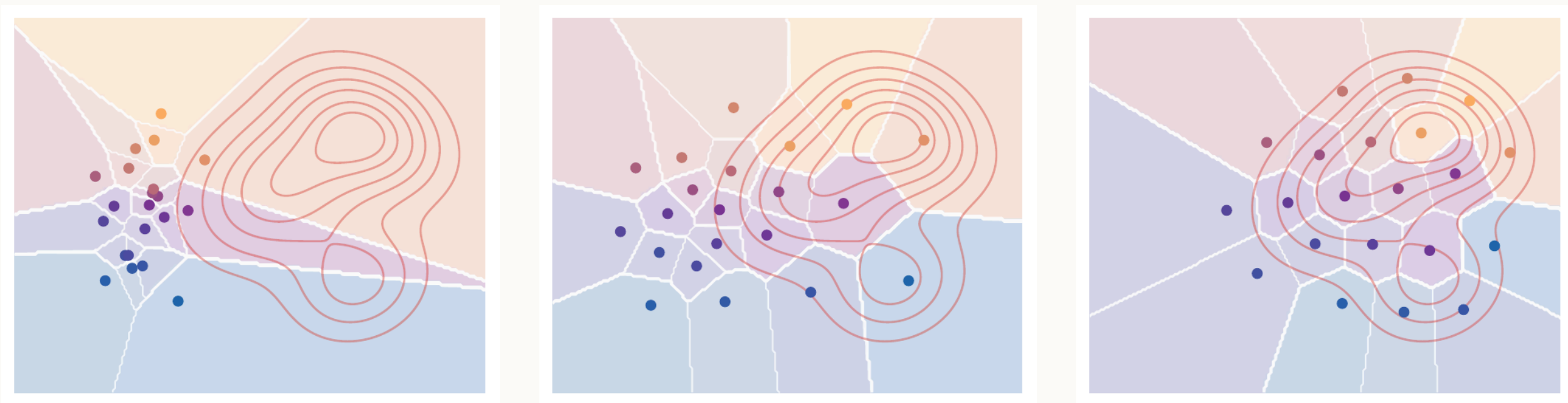
Output: a centroidal Voronoi tessellation.

Live panel: Lloyd iterations



The iterations alternate cell assignment and centroid update.

Lloyd iterations



Lloyd descent is simple and robust, but it is nonconvex and can converge to local minima.

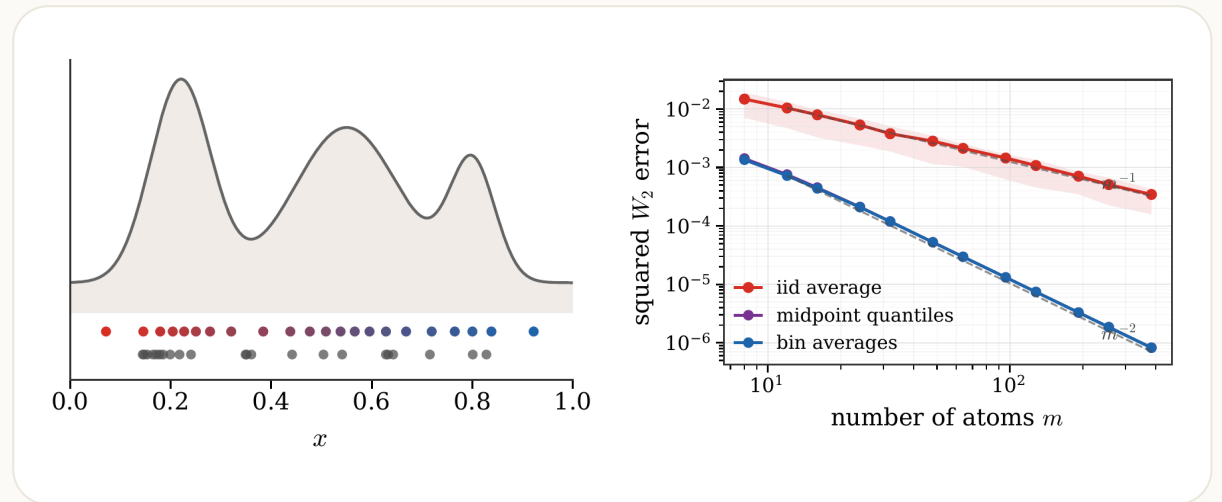
Quantization on the line

In one dimension, quantiles give explicit optimal placements for equal masses. This is much better than random sampling for W_2 approximation.

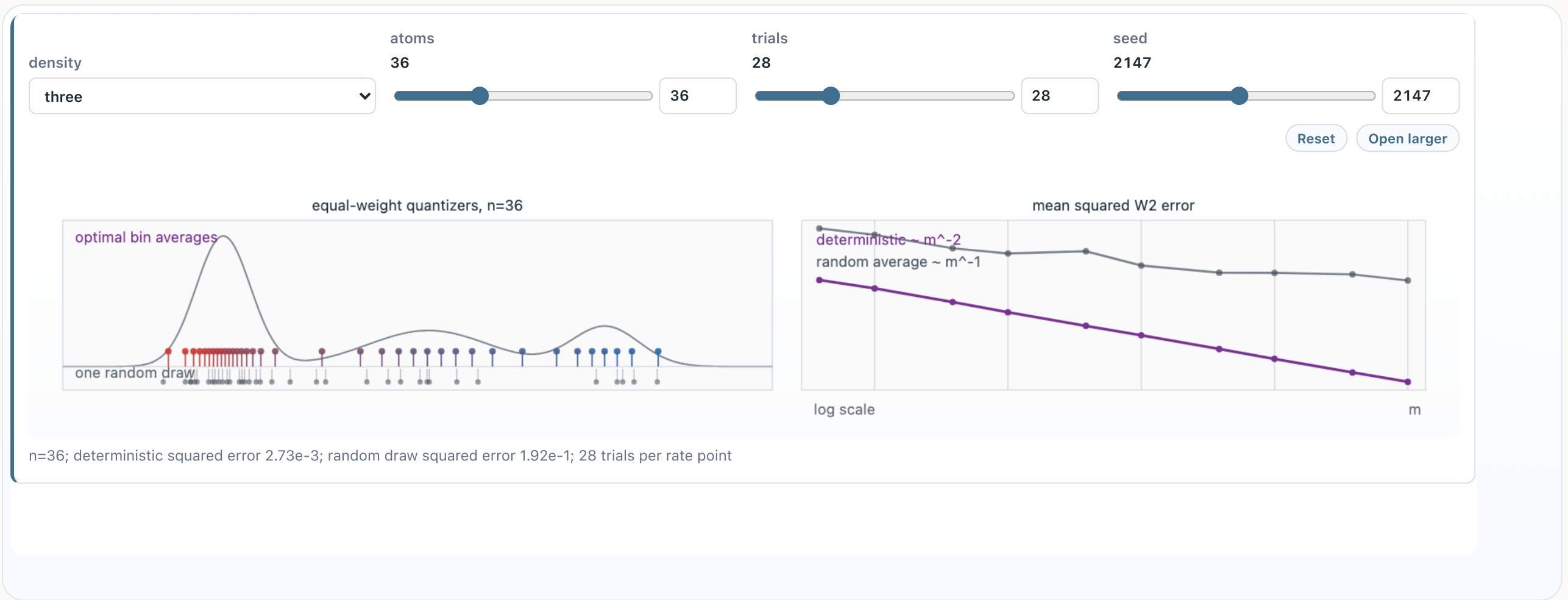
For α with quantile F_α^{-1} ,

$$y_j = F_\alpha^{-1}\left(\frac{j - \frac{1}{2}}{m}\right)$$

is the canonical equal-mass placement.



Live panel: quantization rates

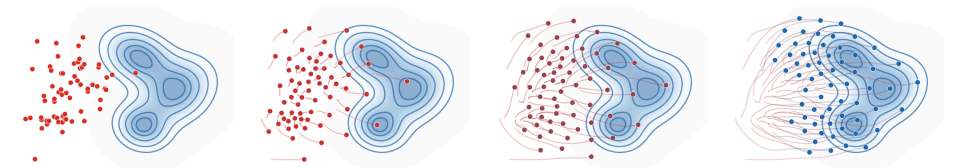


Optimal quantiles and random samples have visibly different convergence behavior.

Relaxed Lloyd flow

A slowed-down Lloyd update can be viewed as a continuous-time descent on the quantization energy in configuration space. For small relaxation parameter η ,

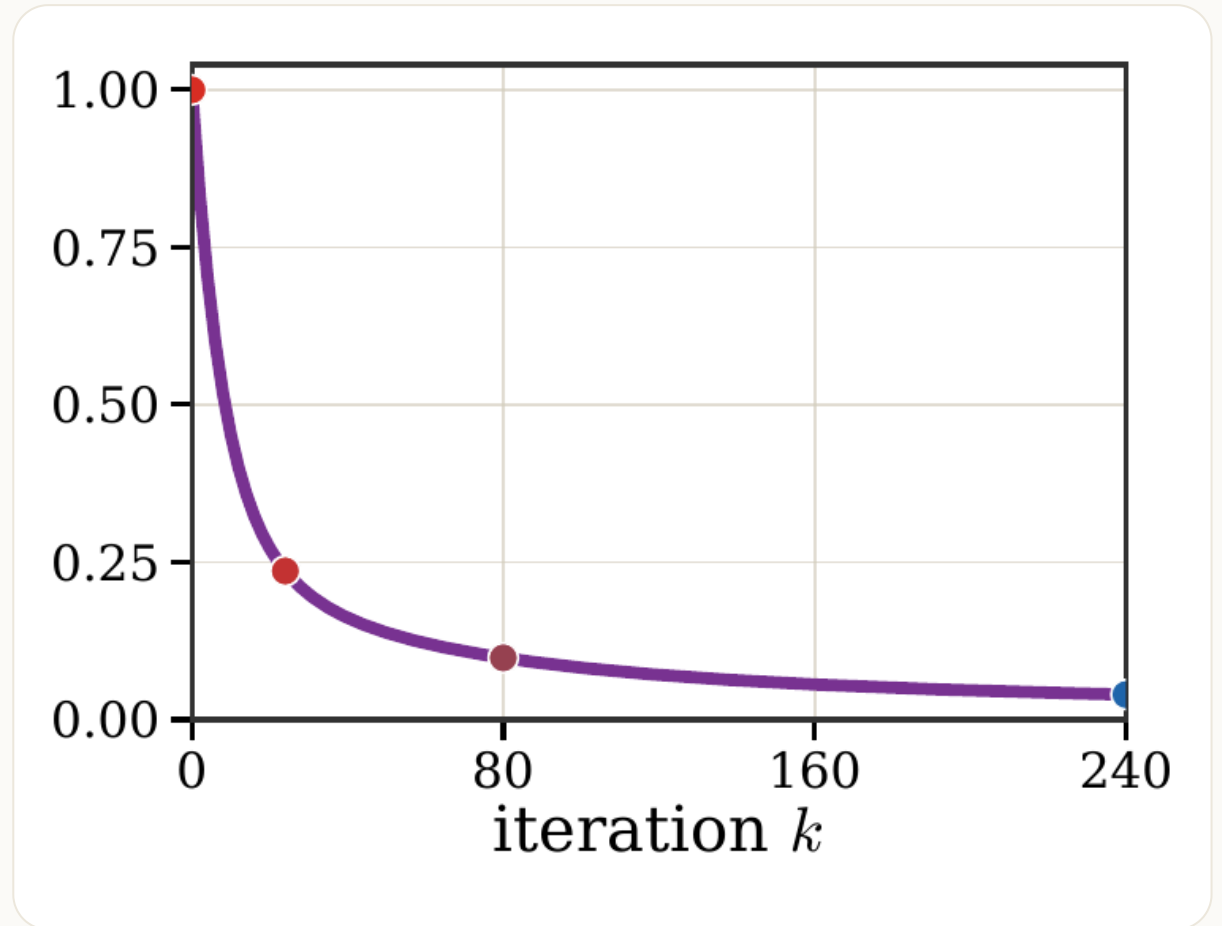
$$y_j^{k+1} = (1 - \eta)y_j^k + \eta \text{centroid}(V_j(Y^k)).$$



Sites move toward a quantized representation of the target density.

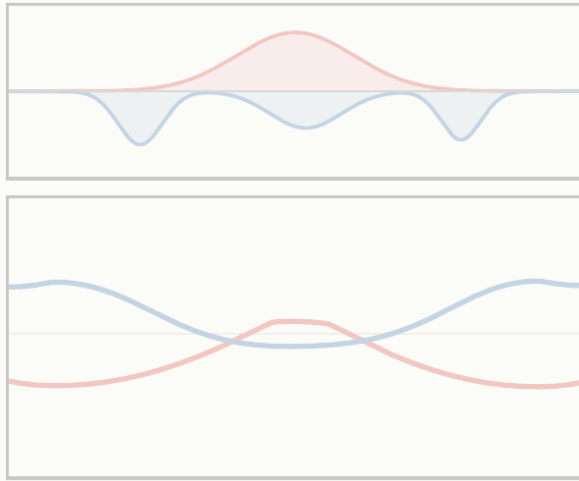
Energy decay

The energy need not be globally convex, but standard Lloyd updates decrease it at each alternating minimization step.

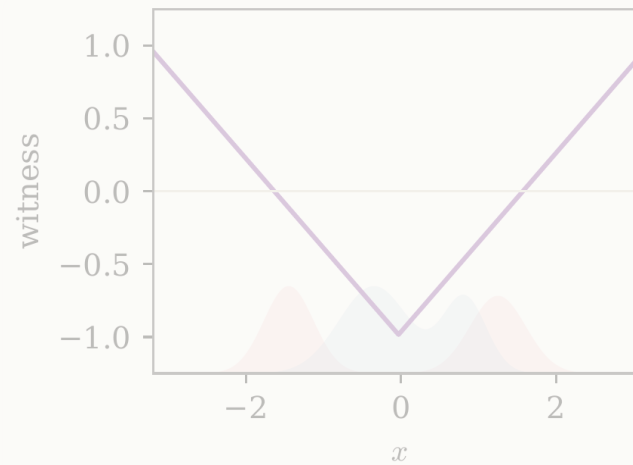


Roadmap: links to learning

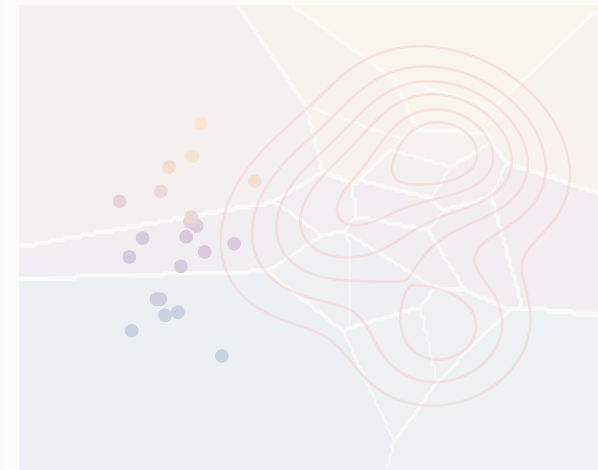
1. Duality



2. Wasserstein-1 and IPMs



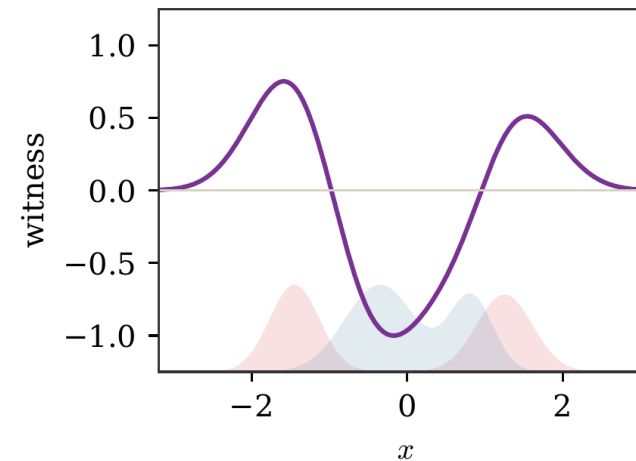
3. Semi-discrete OT



4. Quantization



5. Learning links



Dual potentials as discriminators

The W_1 dual is the mathematical core of Wasserstein GANs, while other IPMs replace the Lipschitz ball by another function class.

TV

bounded witnesses

Wasserstein-1

Lipschitz witnesses

MMD

RKHS witnesses

Semi-discrete maps in applications

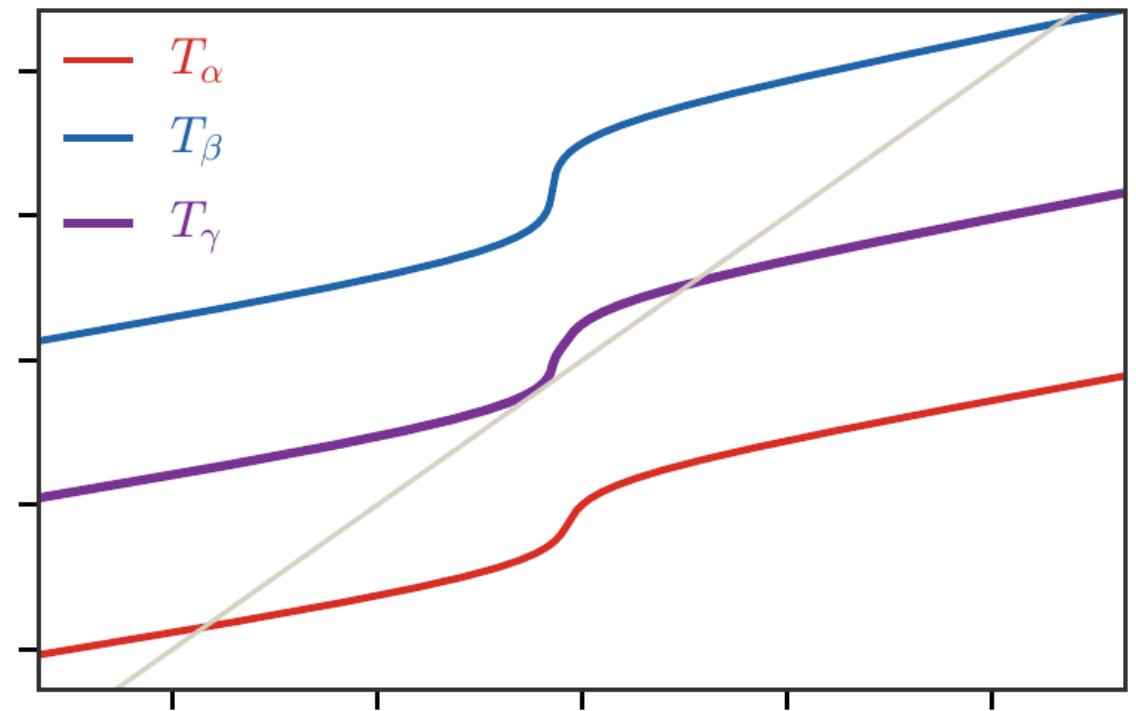
Semi-discrete OT appears when a continuous distribution is represented by a finite set of atoms:

- mesh generation and remeshing,
- color and texture processing,
- sampling and quantization,
- clustering with geometric costs,
- scalable approximations of continuous OT.

The computational object is a partition, not a dense coupling.

Linearized OT preview

Dual and semi-discrete ideas also feed linearized OT: fix a reference measure and embed nearby measures by displacement fields.



Live panel: linearized optimal transport

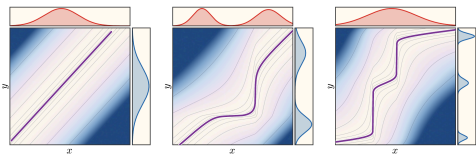


Linear OT turns transport maps into Euclidean coordinates for downstream analysis.

Summary

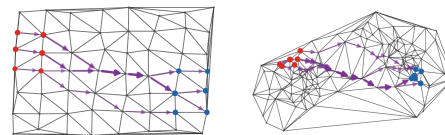
What this deck establishes.

Duality turns transport into potentials. For W_1 , these are Lipschitz witnesses. In the semi-discrete quadratic case, they are Laguerre weights defining a power diagram. Quantization and Lloyd algorithms reuse this geometry.



Duality

certificates and gradients



Wasserstein-1

Lipschitz witnesses



Laguerre

cells and quantization

Next deck

The next presentation focuses on dynamic optimal transport and flows over probability measures.

[Open deck 4: Dynamic Flows →](#)

